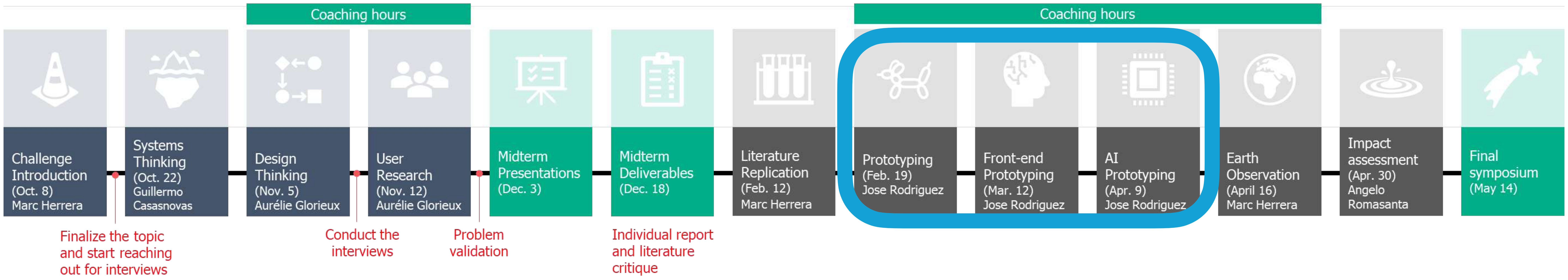
Prototyping Tools & Techniques

Jose A Rodríguez-Serrano | Perspectives on AI, Business & Sustainability

esade

Situational Awareness (I)



Situational Awareness (II)

Innovation projects, such as digital product development, or R&D projects, tend to exhibit four general phases:

	Ideation	Research	Prototyping	Evaluation
Examples	*Design Thinking* *Systems Thinking* *Persona analysis* *Working backwards* *(Powerpoint…)* ~~*Generative AI*~~	*Literature review* *Market research* *User research* *Literature replication*	This part	*Surveys* *User Interviews* *Quantitative metrics* *Usability benchmark*

Prototyping

Prototyping assumes ideation and research are done*. It represents the process of going from the idea to a **tangible demonstrator**.

L8. **Prototyping Tools and Techniques**

Main concepts and theory of prototyping, how all this is challenged by AI.

L9. **Front-end prototyping**

Tools & Ecosystem to create interactive front-ends.

L10. **Prototyping AI Applications**

Tools and best practices for prototyping with data and AI.

(*) Some prototyping techniques are used in ideation phases or prototyping may overlap with ideation.

Prototyping

This part of the course **is** about:

- Learning the theory and mindsets of prototyping as a way of “thinking by doing”.

- Understand specificities of data / AI prototypes.

- Understanding the python ecosystem for prototyping AI applications.

The course **is not** about:

- Agile methodologies / personal productivity with AI / n8n / vibe coding / openclaw…

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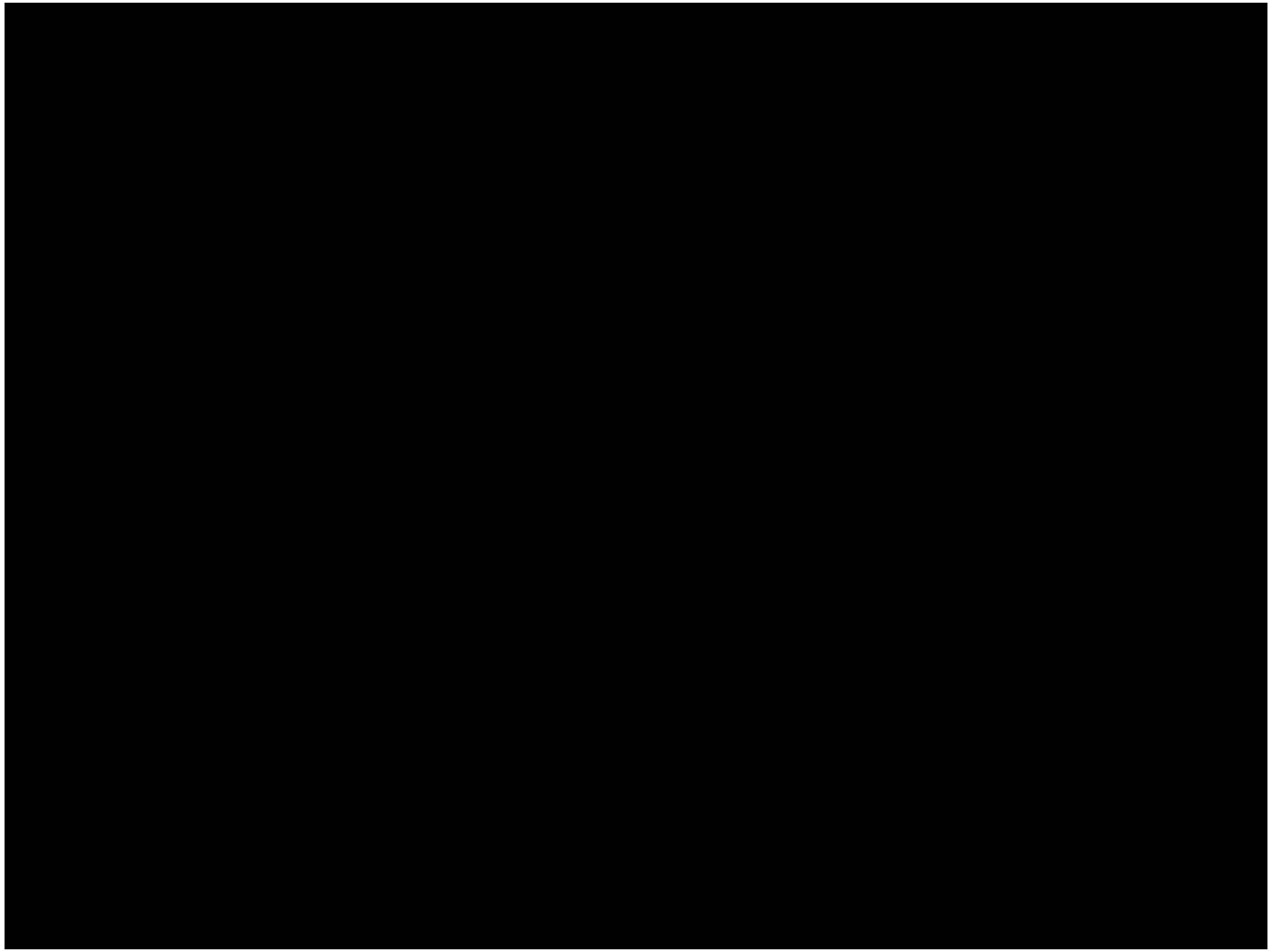
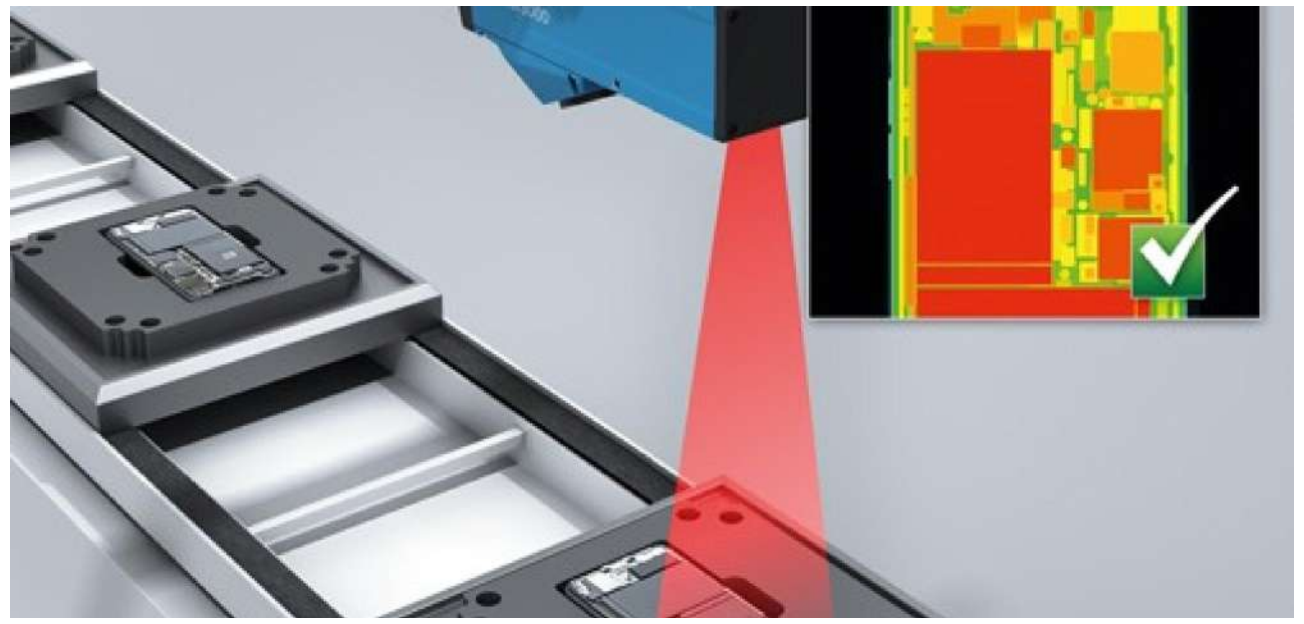
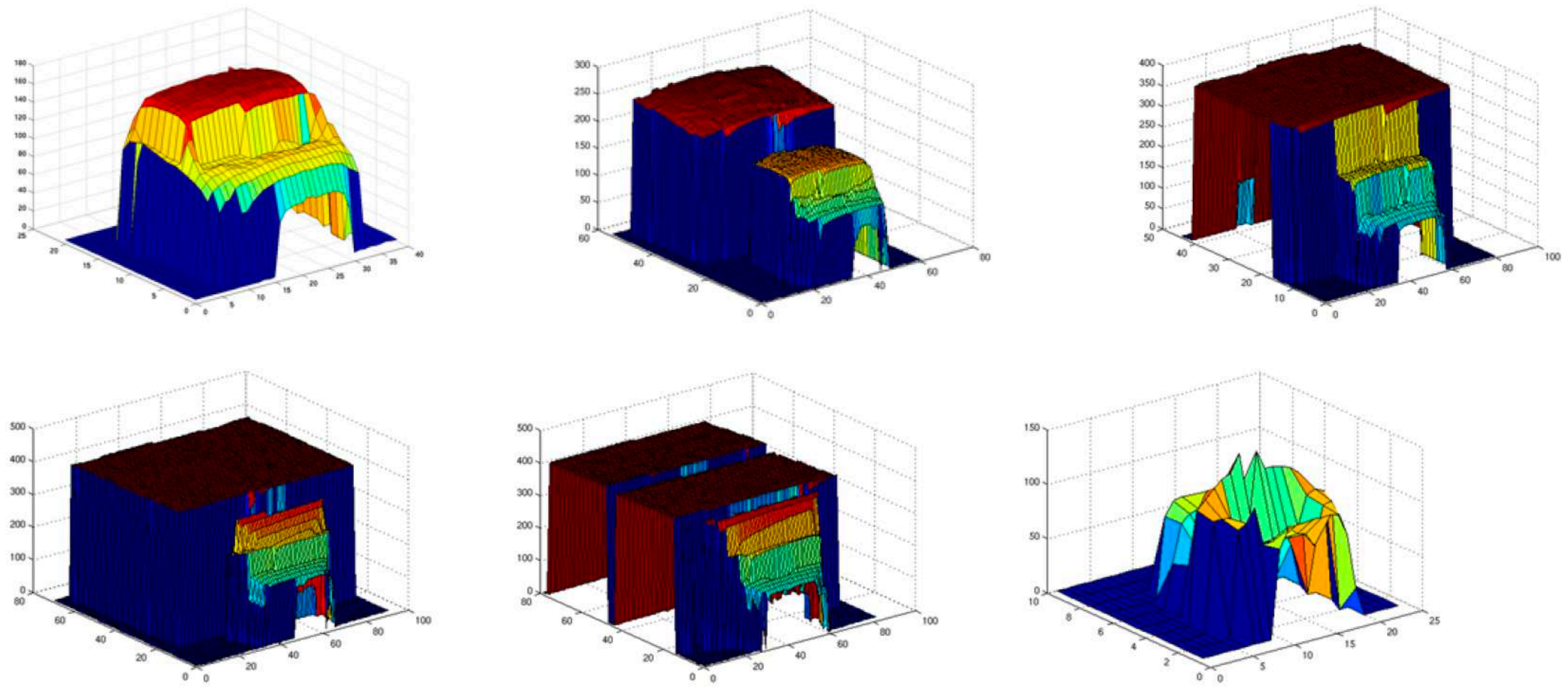
Some prototypes

First prototypes



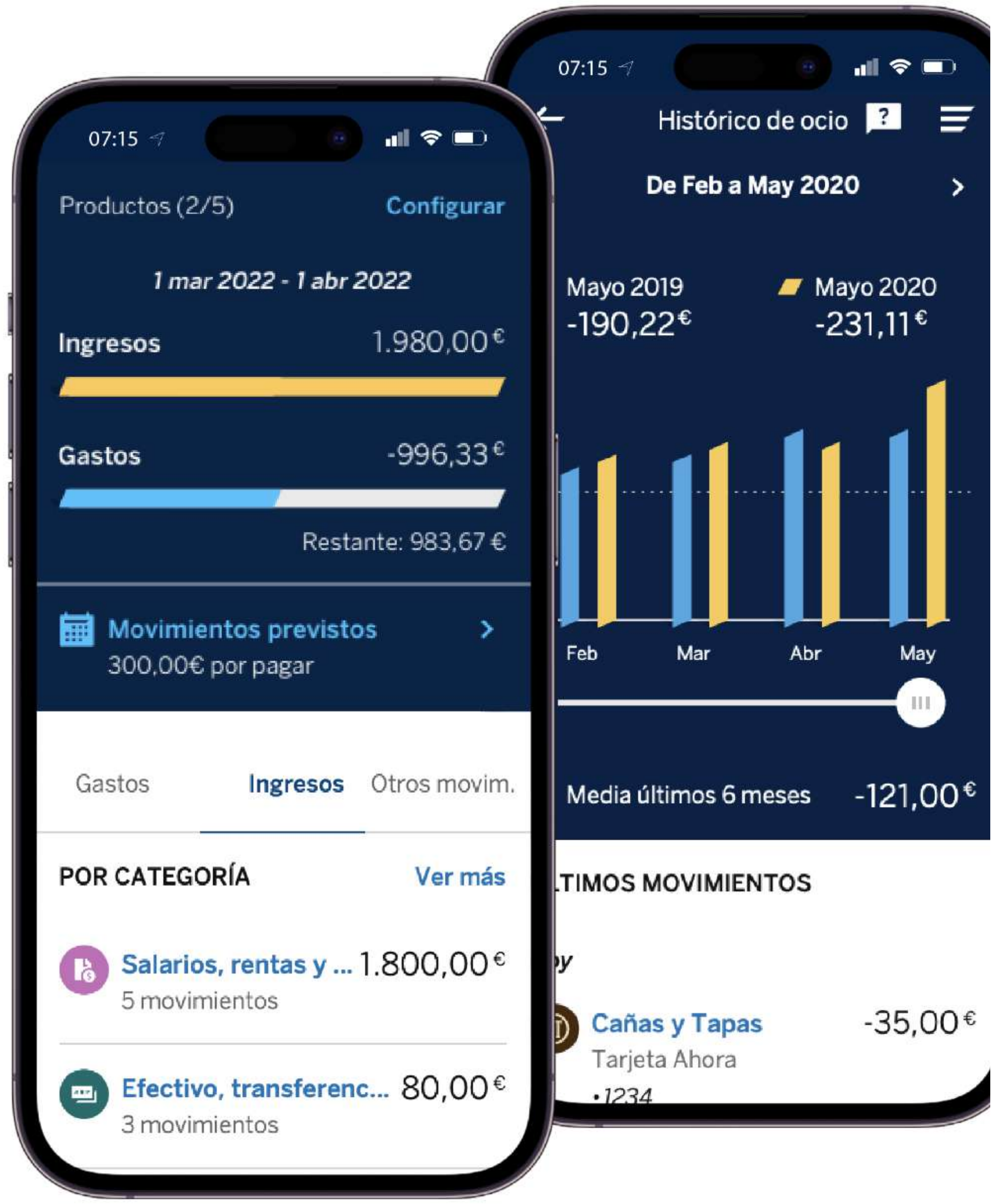
First prototypes

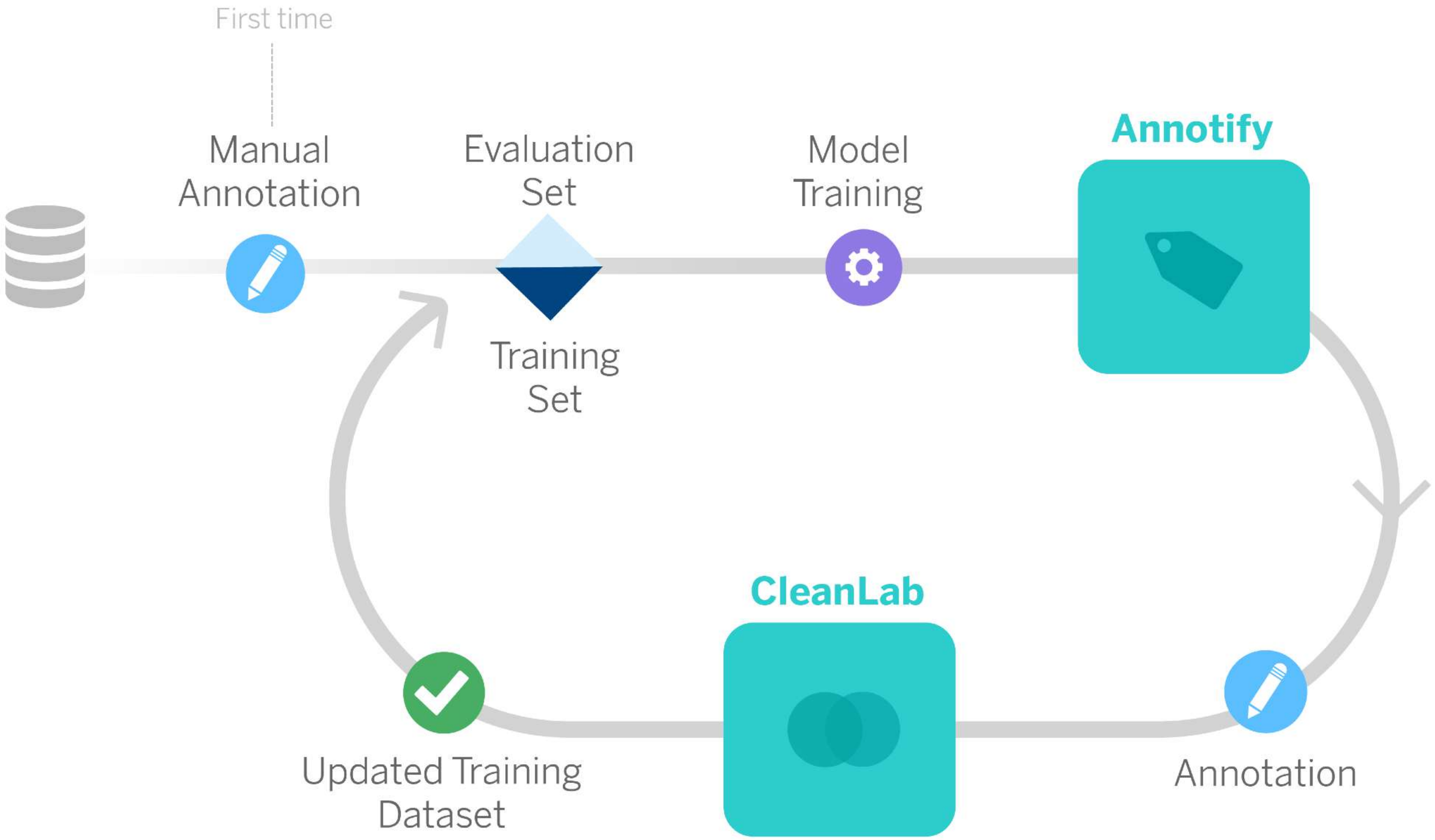




Sandhawalia et al., Vehicle Type Classification from Laser Scanner Profiles: A Benchmark of Feature Descriptors, IEEE Intelligent Transport Systems, 2013.

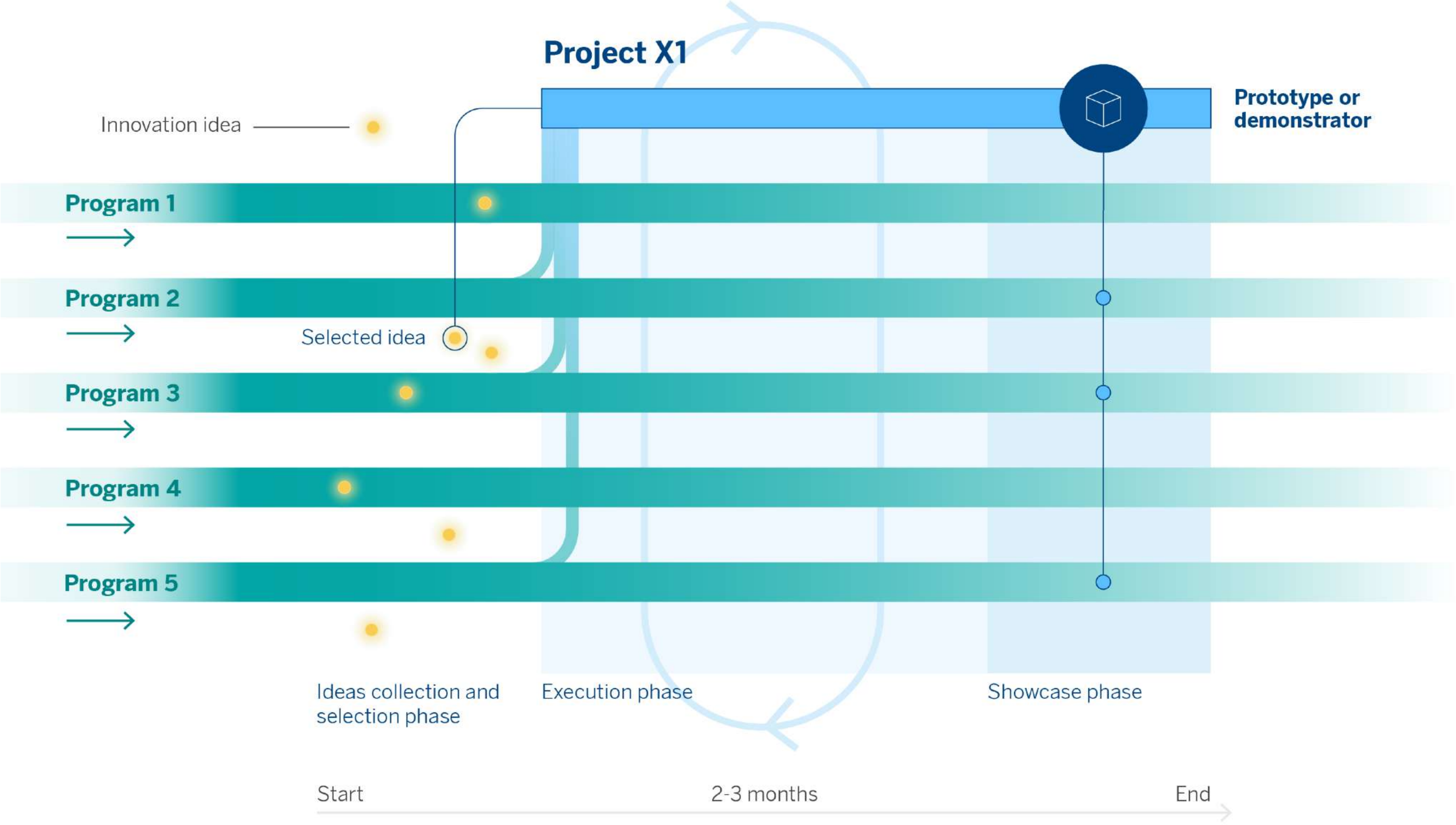
Latest prototypes





https://www.bbvaaifactory.com/money-talks-how-ai-helps-us-classify-our-expenses-and-income/

A prototyping process



Source: About Program X and why we have been awarded for this initiative, BBVA AI Factory

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A bit of prototyping theory (and mindset)

What is a prototype

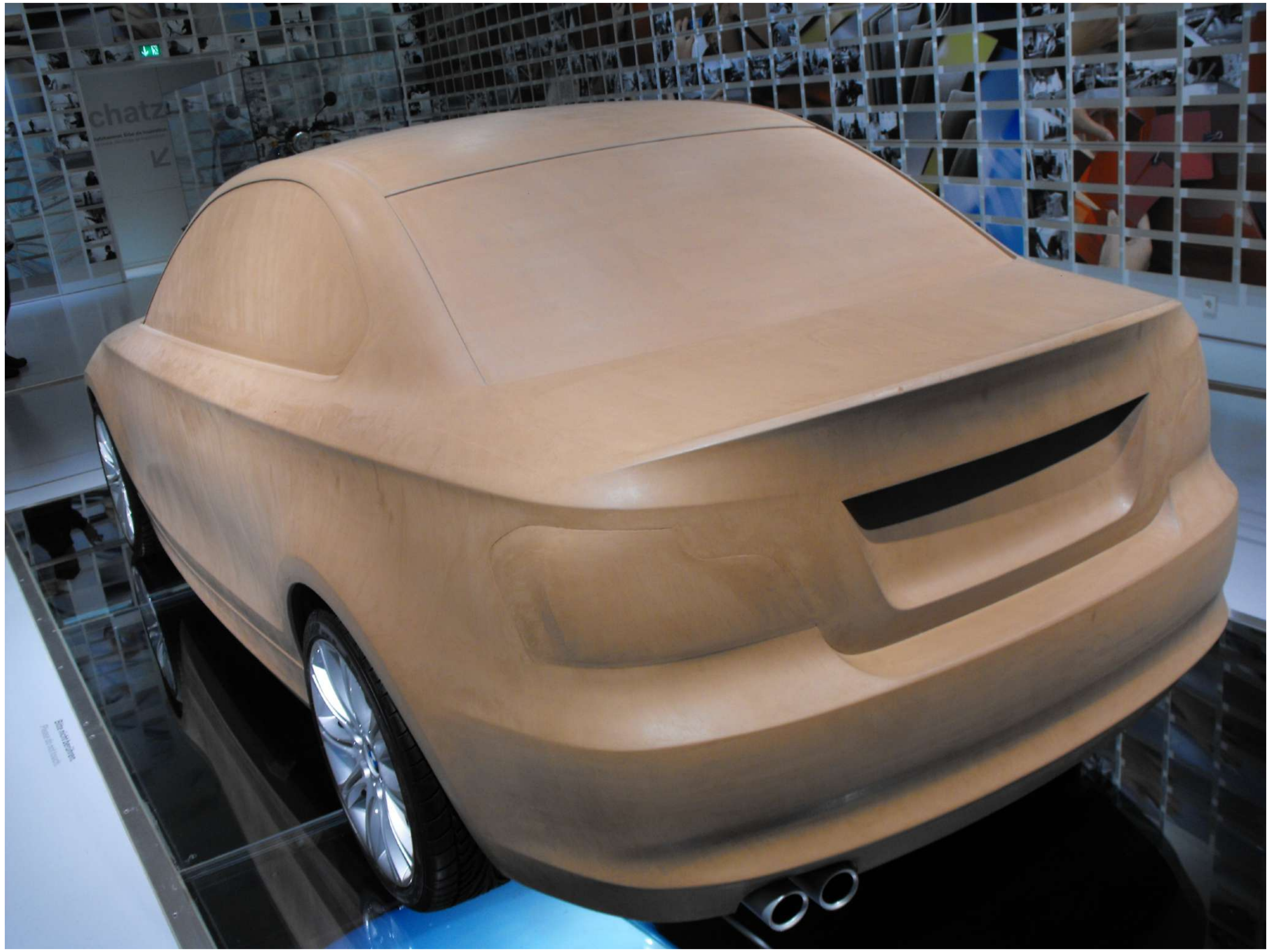
“Prototype” may mean something different to different audiences.

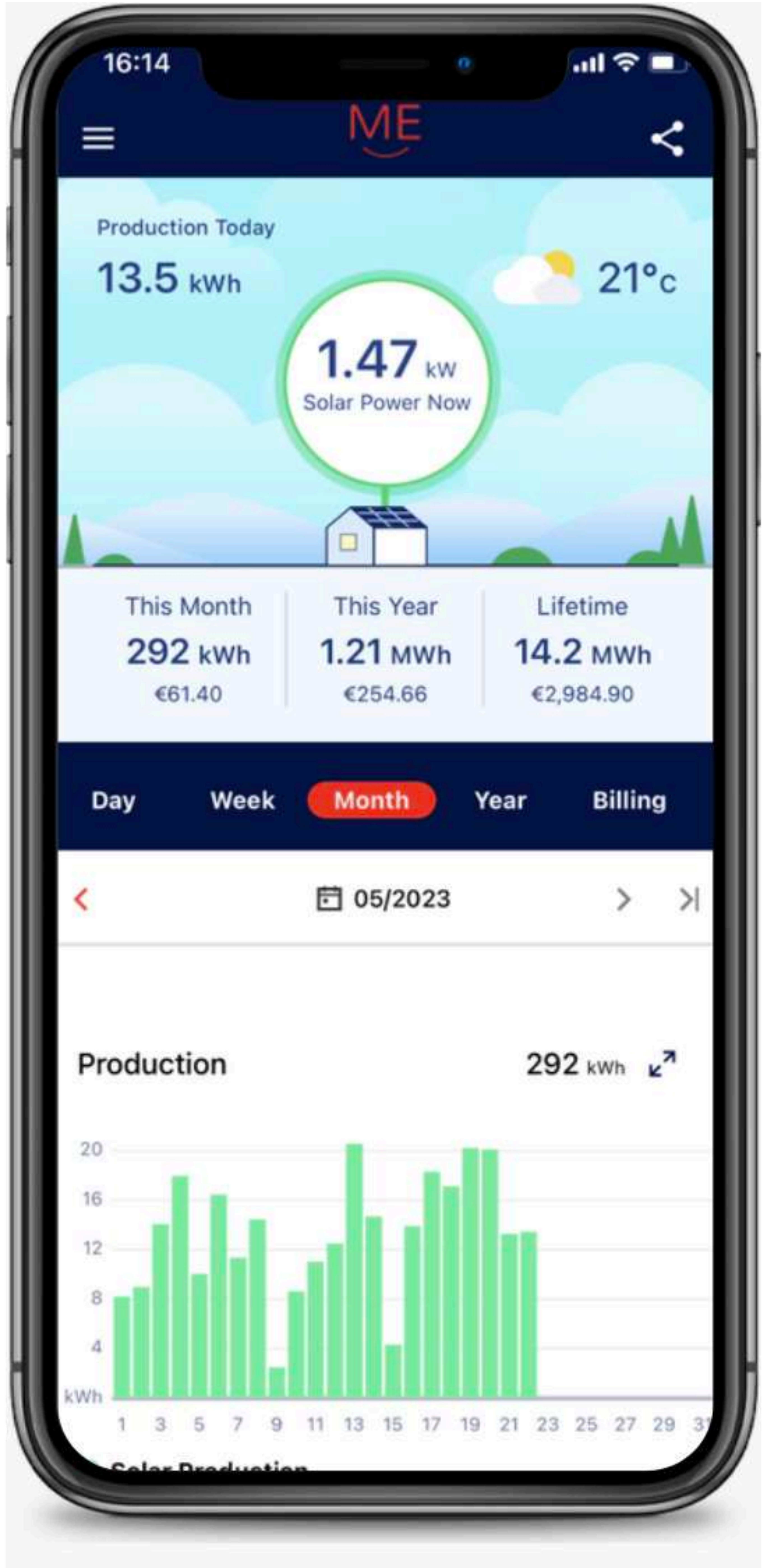
- An MIE student may call an Figma screen simulation a “prototype”.

- An interaction designer may call a hand-drawn storyboard a “prototype”.

- A programmer may call a test program a “prototype”.

- An industrial designer may call a 3D print a “prototype”.




16:14
ME
Production Today
13.5 kWh
21°c
1.47 kW
Solar Power Now
This Month
292 kWh
€61.40
This Year
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€254.66
Lifetime
14.2 MWh
€2,984.90
Day
Week
Month
Year
Billing
05/2023
Production
292 kWh
20
16
12
8
4
kWh
1 3 5 7 9 11 13 15 17 19 21 23 25 27 29

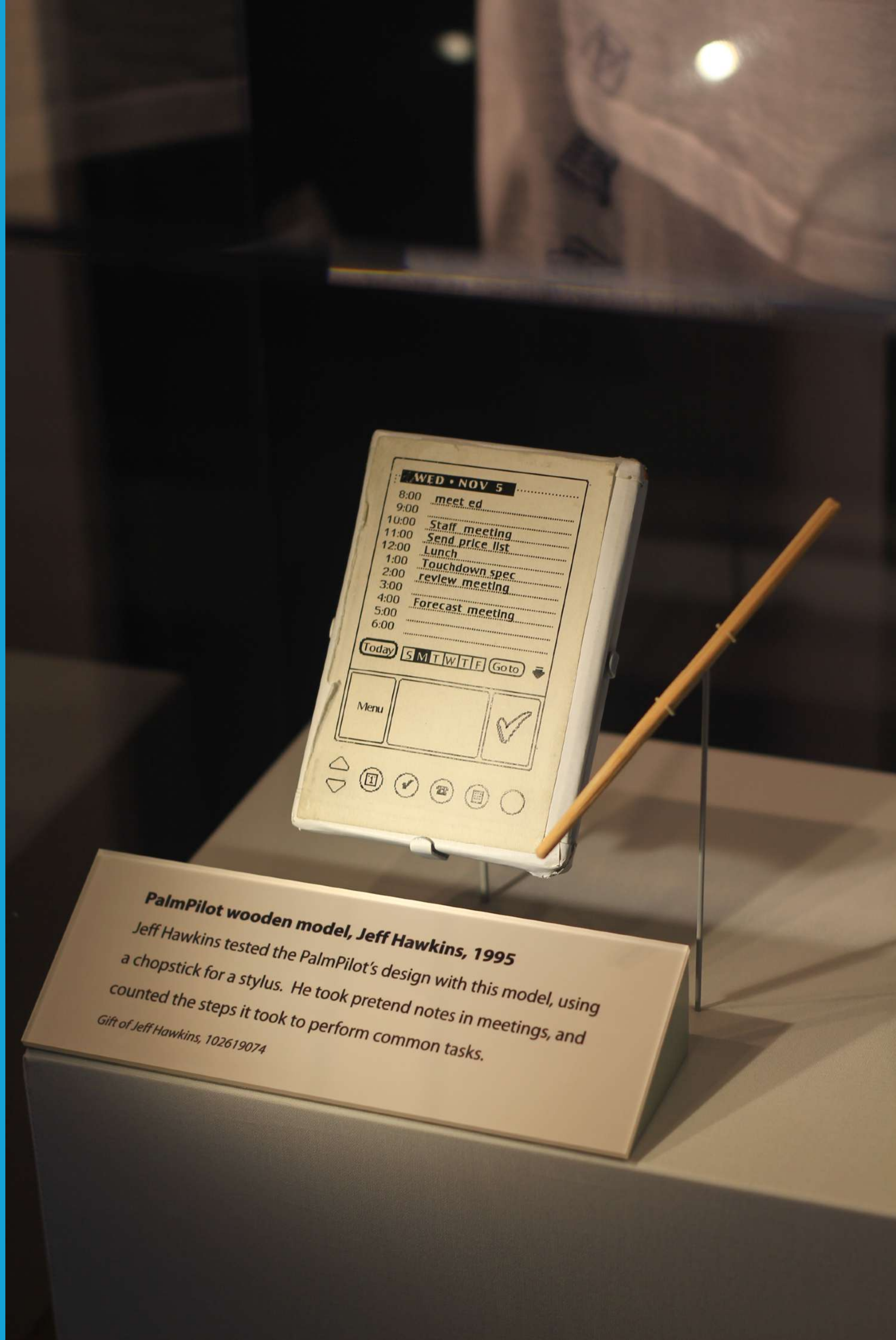
A prototype is a *representation of a design idea* in any medium. It is a core means to *explore* and *express* designs.

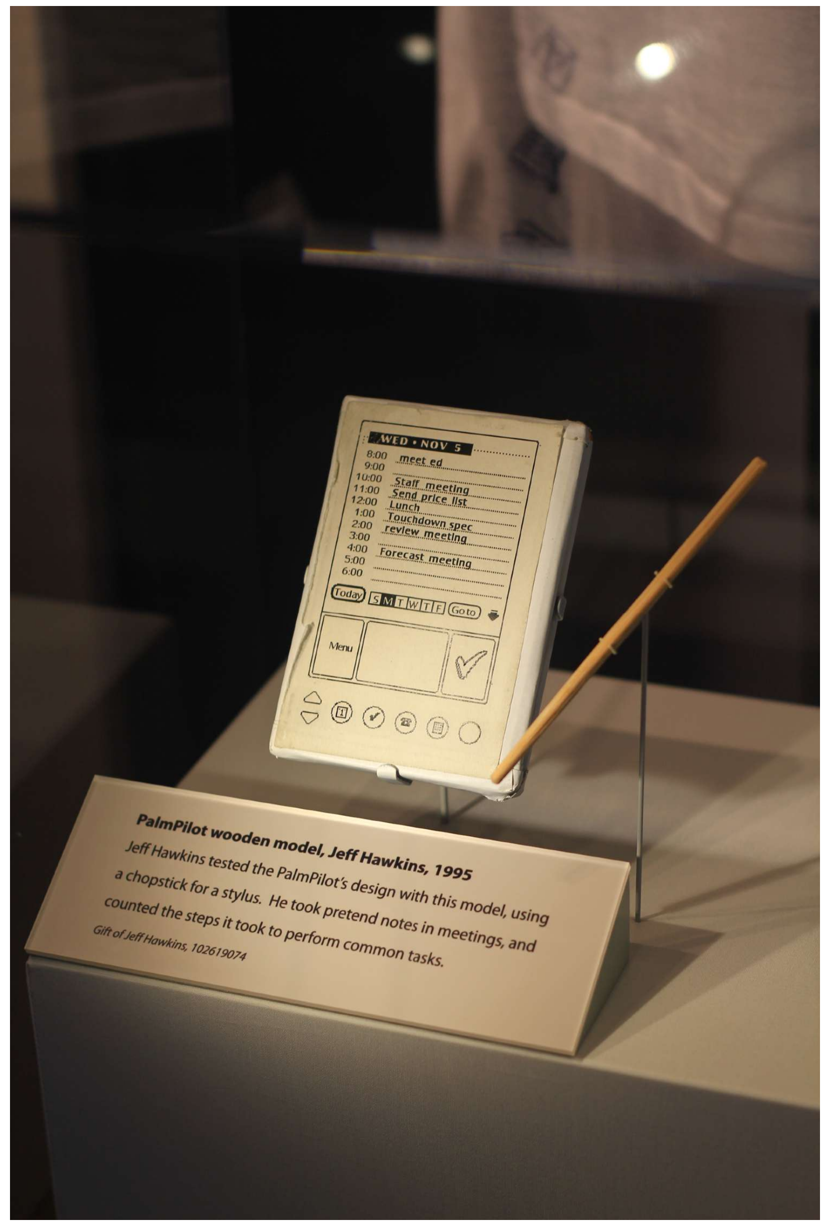
Inspired by**: Stephanie Houde & Charles Hill,** What do prototypes prototype?

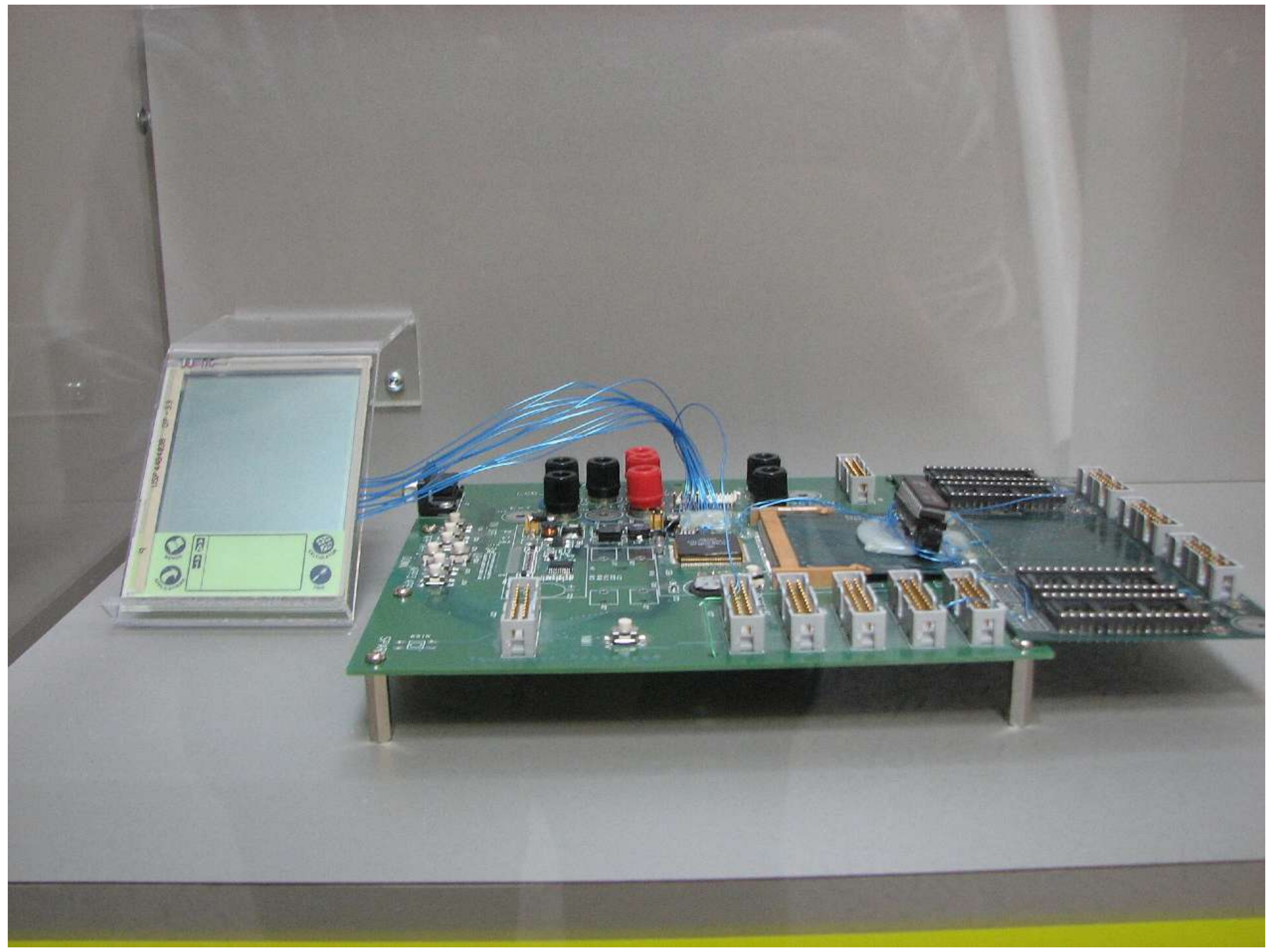
Prototyping is about **making sure we are building the right thing** before starting the actual (and more costly) development work.

Prototyping encompasses **a range of experimental procedures and media** rather than adhering to a singular, fixed methodology. But the basic characteristics are:

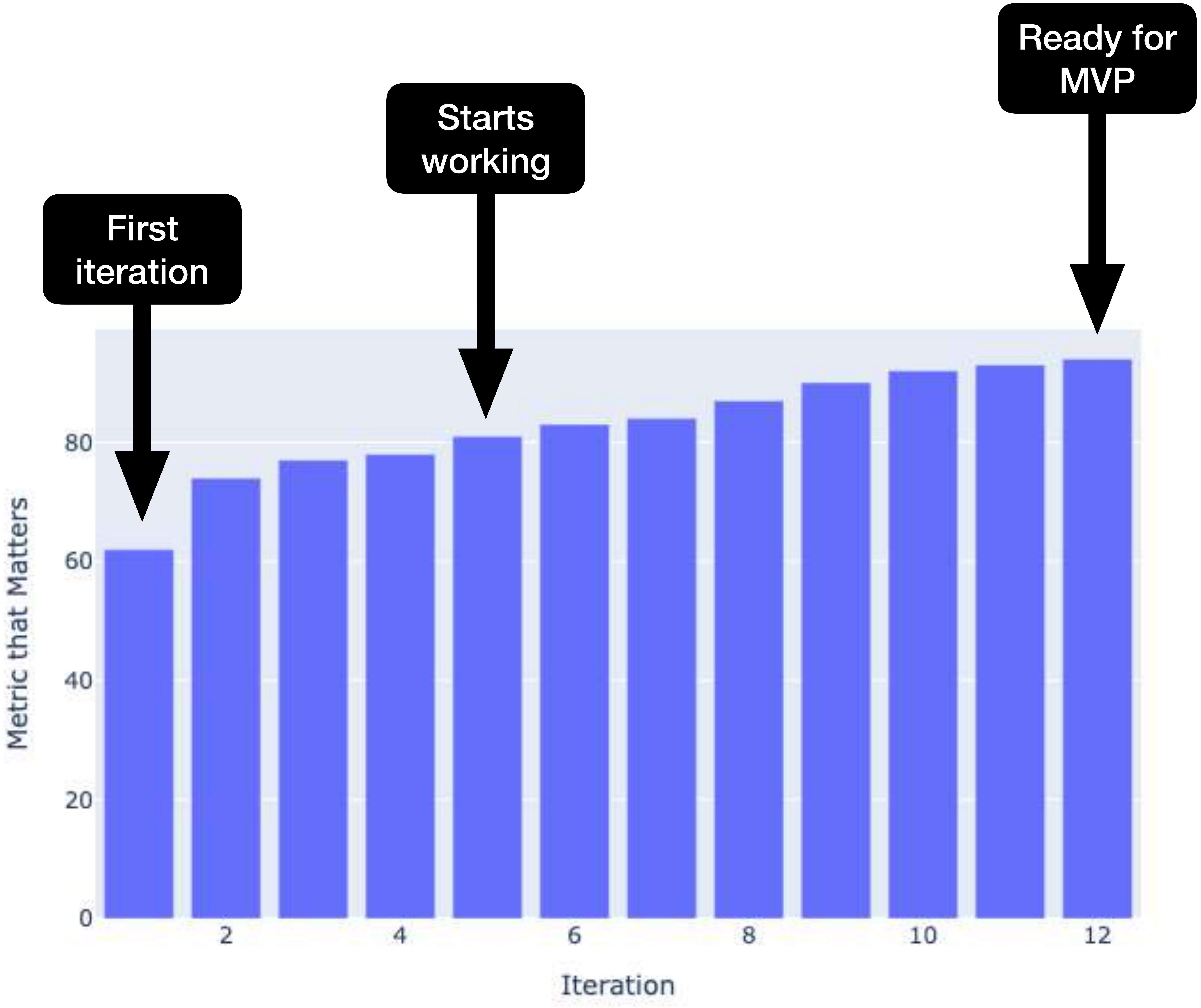
- **Modest scale** and **execution speed:** A prototype should be "lean": employ the minimum resources and time to extract maximum learnings.
- **Iterative refinement:** Each iteration of a prototype answers specific questions or reduces specific uncertainties.
- **Solicitation of feedback:** A prototype allows for exploration and interaction, making it a vehicle to obtain feedback.
- **Low-fidelity:** A prototype is always unfinished and should deliberately look unfinished. This eases the provisioning of feedback.

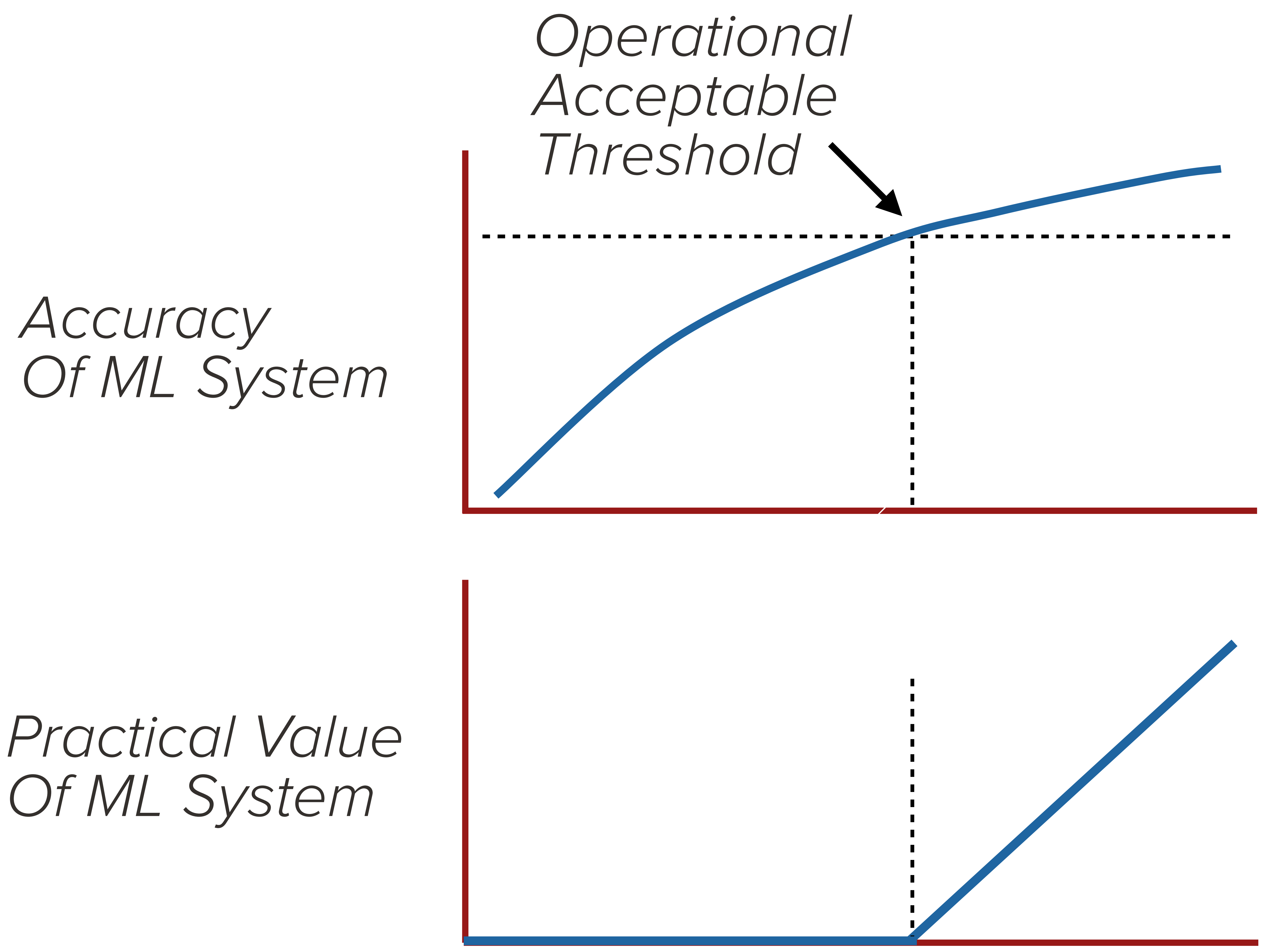


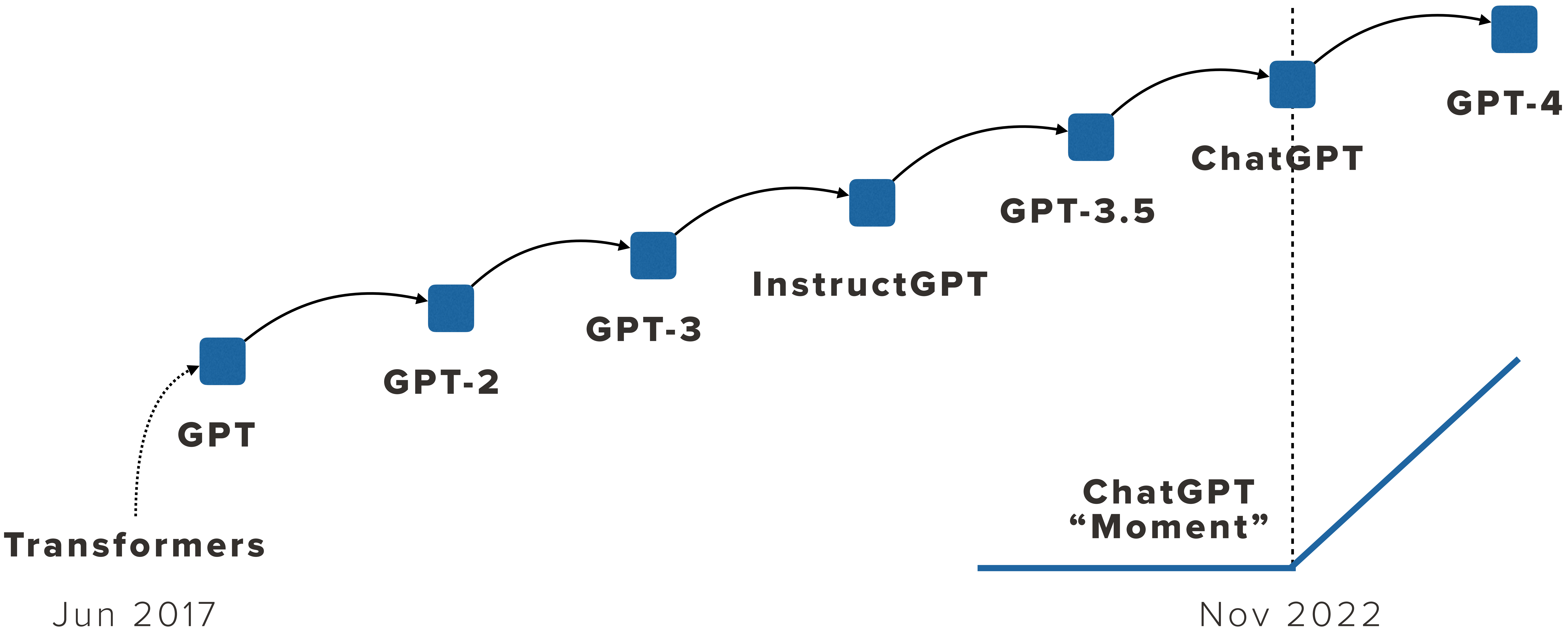

PalmPilot wooden model, Jeff Hawkins, 1995
Jeff Hawkins tested the PalmPilot's design with this model, using a chopstick for a stylus. He took pretend notes in meetings, and counted the steps it took to perform common tasks.
Gift of Jeff Hawkins, 102619074
Menu




palm


First iteration
Starts working
Ready for MVP
Metric that Matters
0
20
40
60
80
2
4
6
8
10
12
Iteration


Operational
Acceptable
Threshold
Accuracy
Of ML System
Practical Value
Of ML System


GPT-4
ChatGPT
GPT-3.5
InstructGPT
GPT-3
GPT-2
GPT
ChatGPT
“Moment”
Transformers
Jun 2017
Nov 2022

Reasons for prototyping

Better showcase your idea

A simple demo of your product can get your team & stakeholders engaged.

Discard bad ideas early on

A prototype is a natural tool to "Fail fast" & de-risk

Force you to think about details

By actually building the think you come across questions and details not available before.

Explores "the neighborhood"

A prototype fosters exploring where no one else is looking.

Supports decision-making

Making something specific enables clearer decisions.

See also:

“Sometimes, well-thought-out proposals backed by extensive research and data will fail to convince decision-makers. But a demo of a simple prototype will get them excited and ready to commit. I’m surprised how often this happens.” — Eugene Yan

Ziyou Yan: How prototyping can help you get buy-in, 2020.

``frequently, [...] a better idea would never have come about were it not for the idea that it replaces''.

Bill Buxton, Sketching User Experiences

“Let the prototype do the deciding”

Ben Holliday: https://benholliday.medium.com/the-prototyping-mindset-320de6ec79c8

"There is no formula. No bucket list. No destination.
There is only learning"

Ha Phan: https://hpdailyrant.medium.com/letter-to-a-young-designer-wax-on-wax-off-b9457dcca568

This has other names…

In the fields of creativity, entrepreneurship, art, digital product development, famous sentences have captured similar mindsets.

Start before you’re ready

Better done than perfect

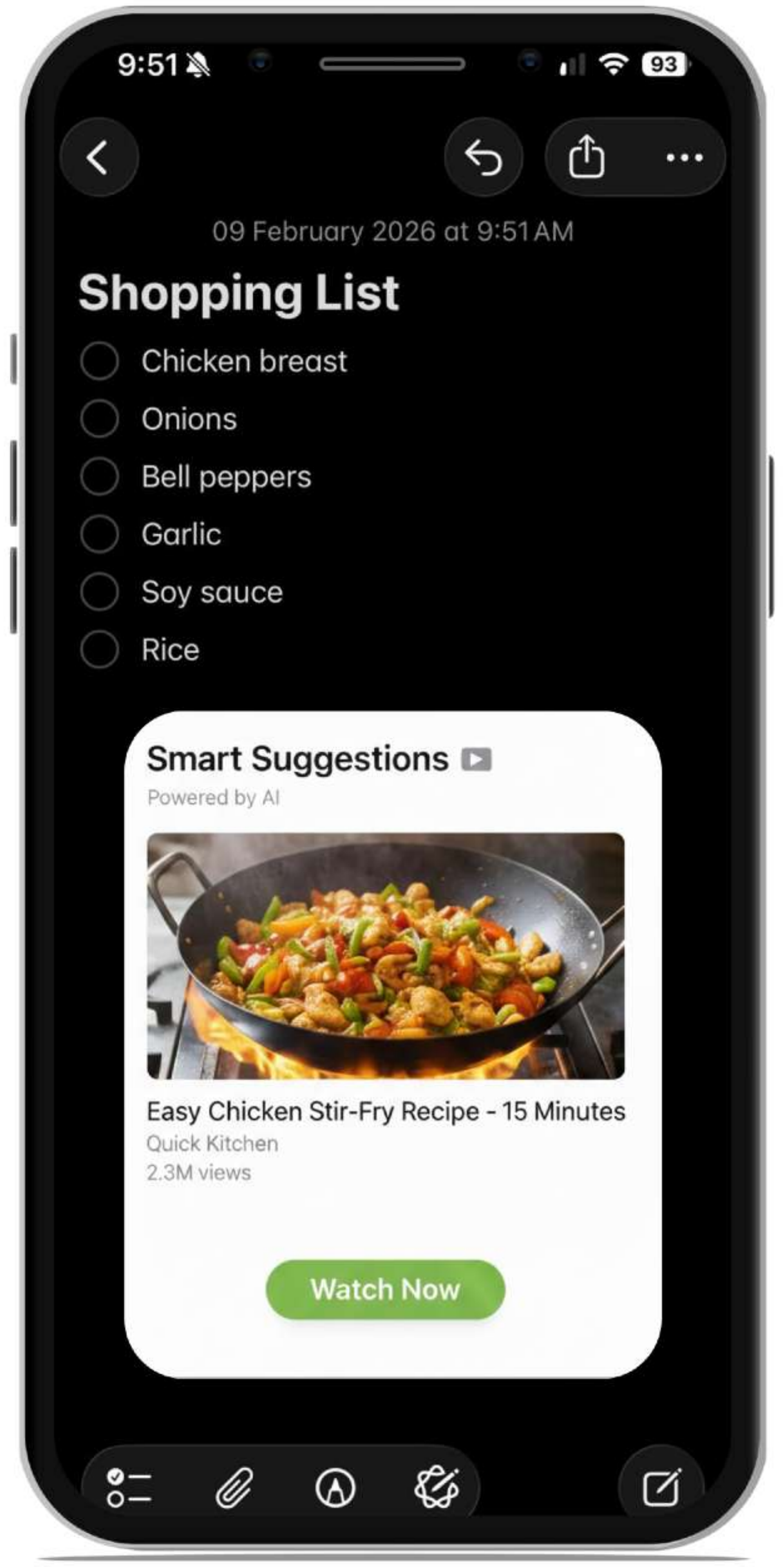
Move fast and break things

Release early, release often

Fail fast

Fake it ‘till you make it

Prototyping case



A notes-taking app with a "Smart Suggestions" feature that prompts a Youtube video related to the note items.

Prototypes should seek answers to the most pressing questions

Lecture 1.4 The Power of Prototyping

Prototyping questions

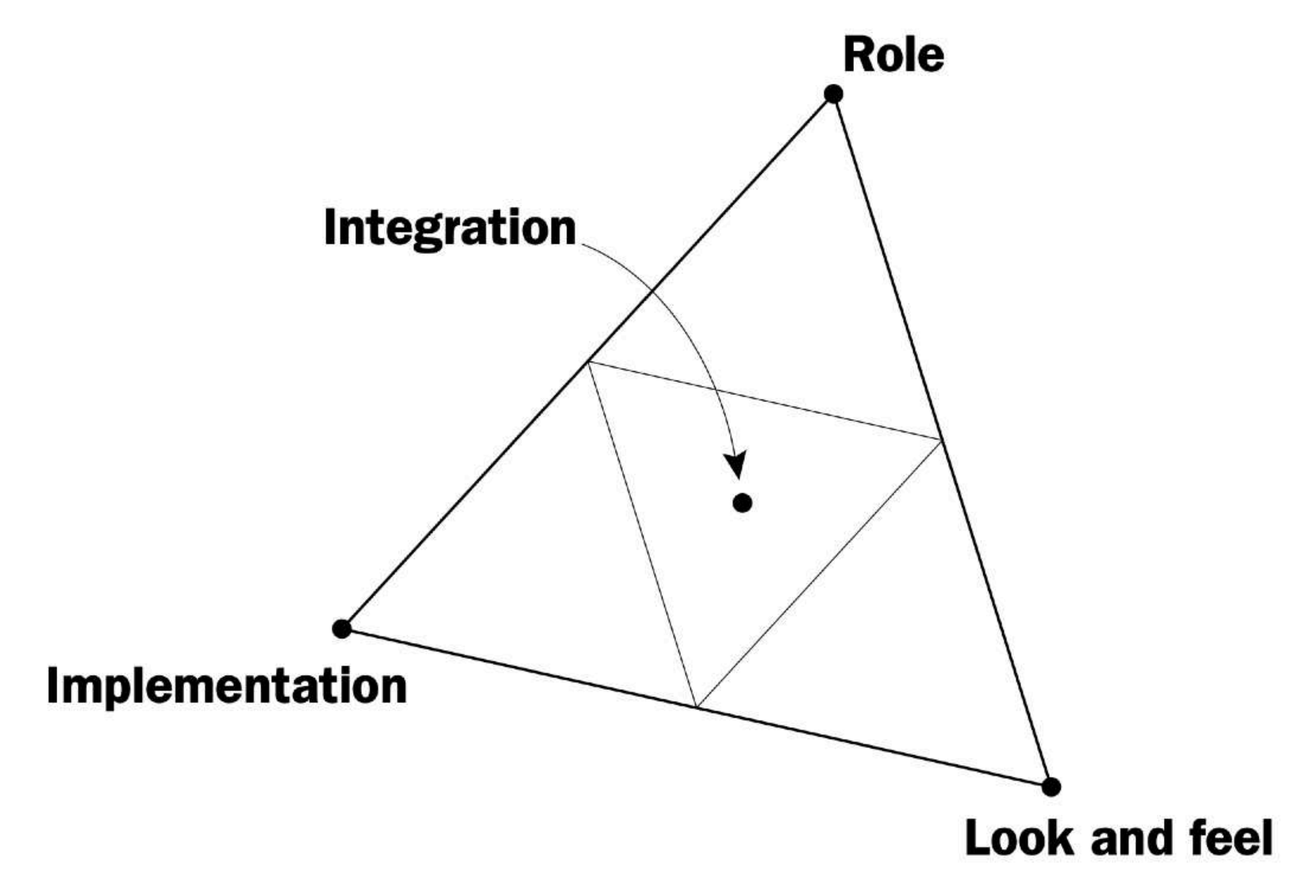
Houde and Hill (2012) classify prototyping questions into four families:

> **Role**: Answer what a product could do for a user.

> **Look and feel**: Explore and demonstrate options for the concrete experience (which includes appearance).

> **Implementation**: Answer technical questions about how a future artifact might actually be made to work

> **Integration**: Represent the complete user experience, bringing together aspects from Role, Look and feel, and Implementation.



Stephanie Houde & Charles Hill
What do prototypes prototype?

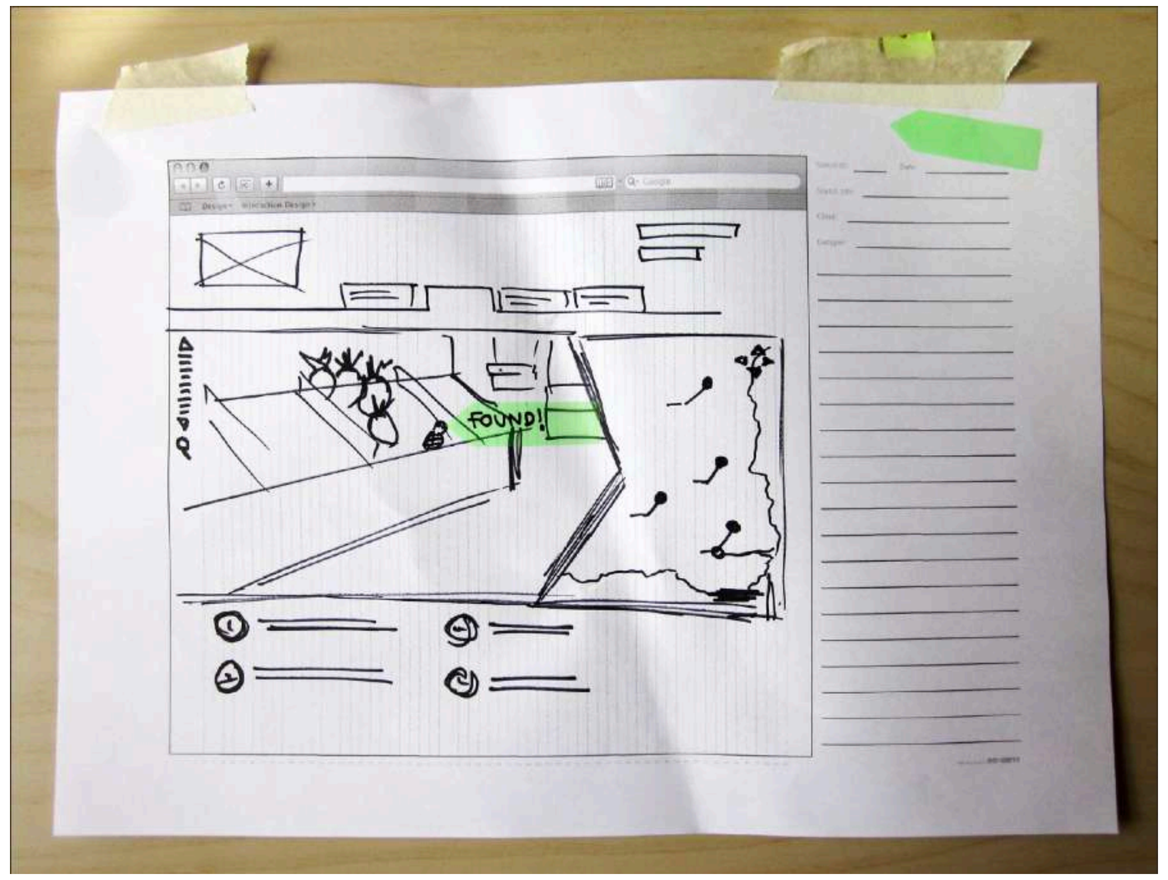

4627
4 6 2 7

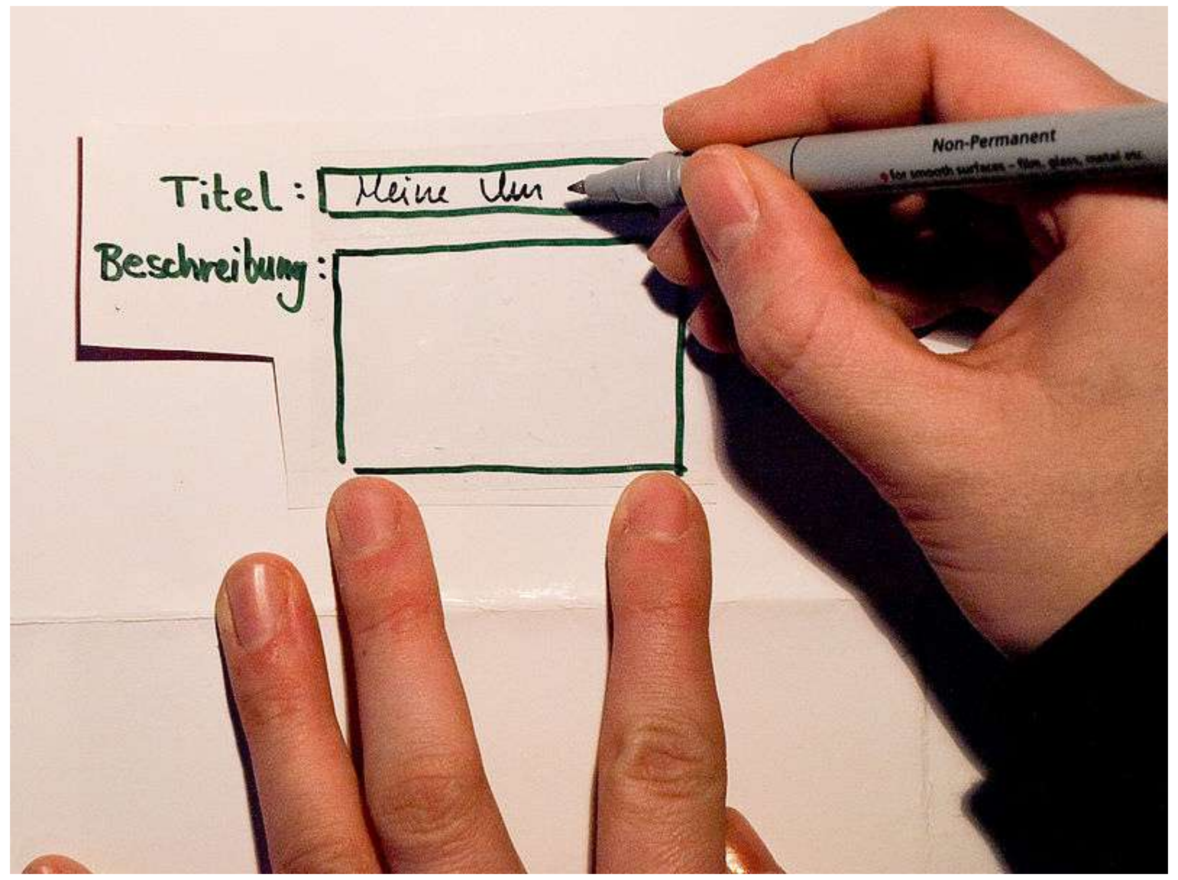
Some (traditional) prototyping strategies

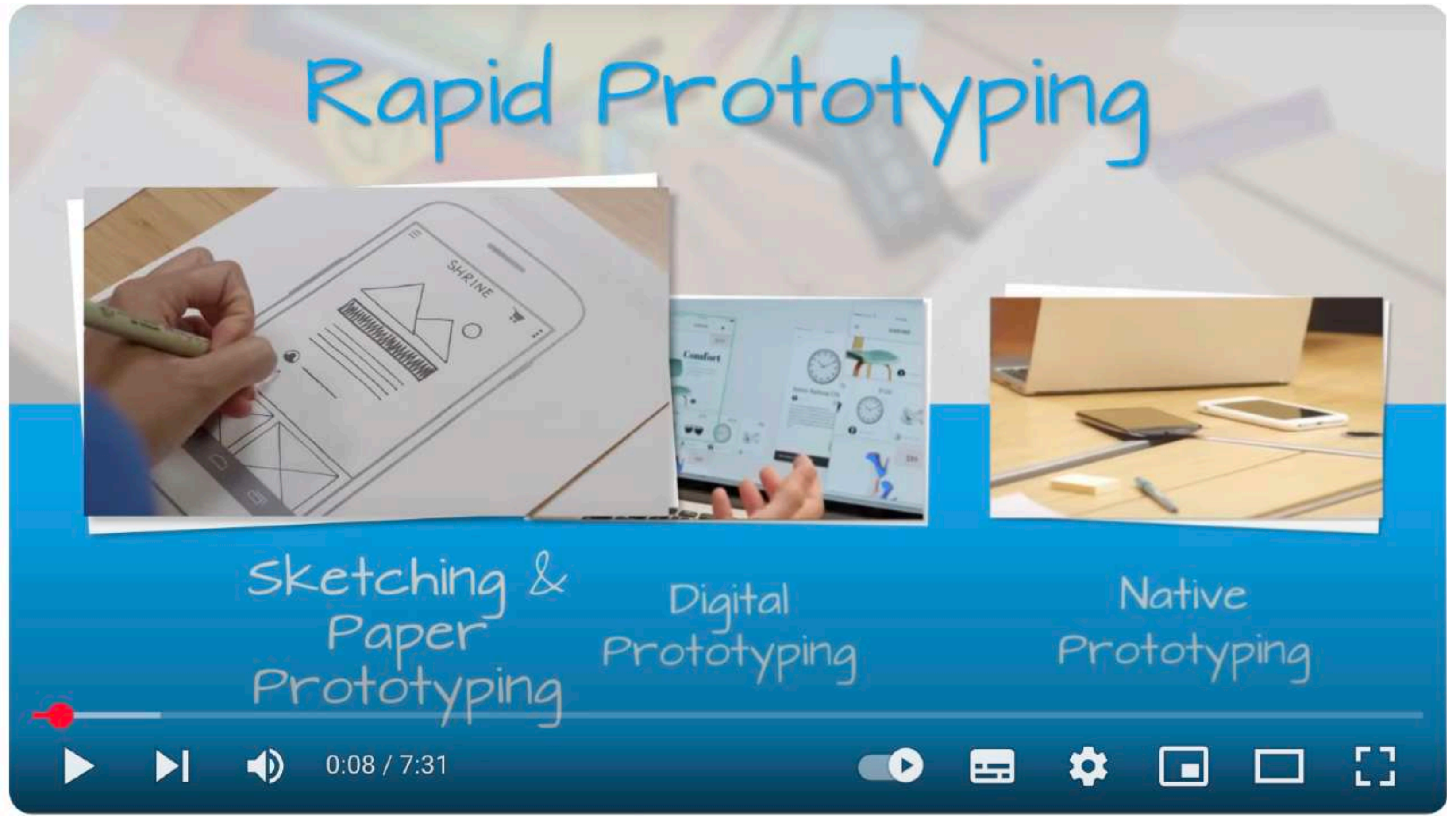
Paper prototyping

It uses printed or drawn materials to represent the layout and interactions of a user interface. It is cheap and useful at the early stages of a design exploration.

Different paper techniques (e.g. post-its) may be used to simulate interactions, and interactions <u>can be recorded to simulate real app behaviour</u>.

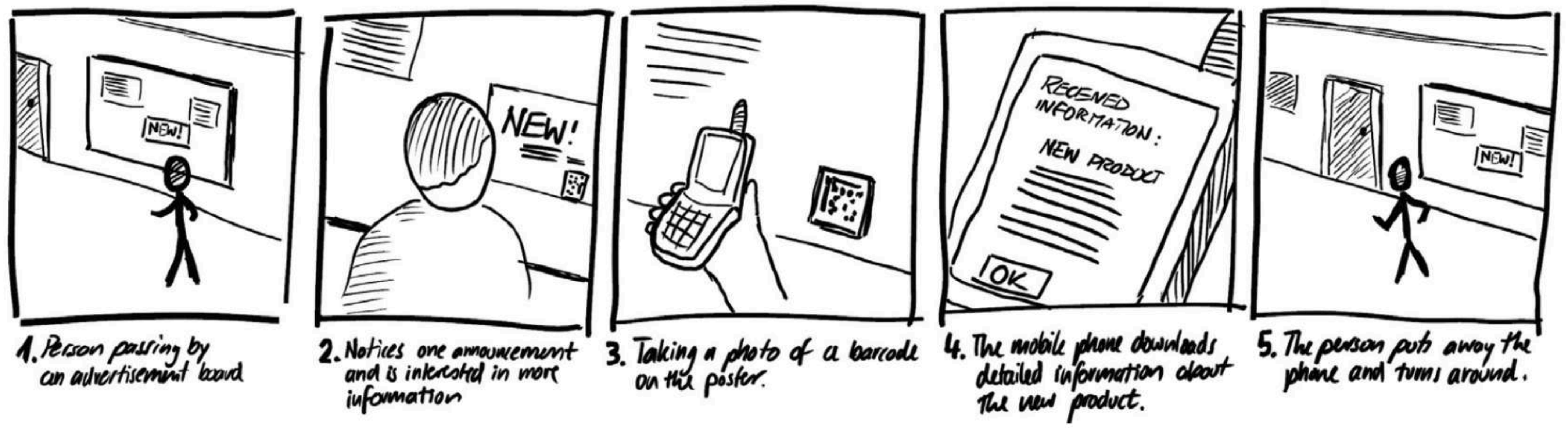






Storyboarding

- A storyboard depicts a “journey”.
- Used by designers to anticipate how users will interact with a product over time.
- Capability to outline user actions at each stage (but also allows representing thoughts or emotions).



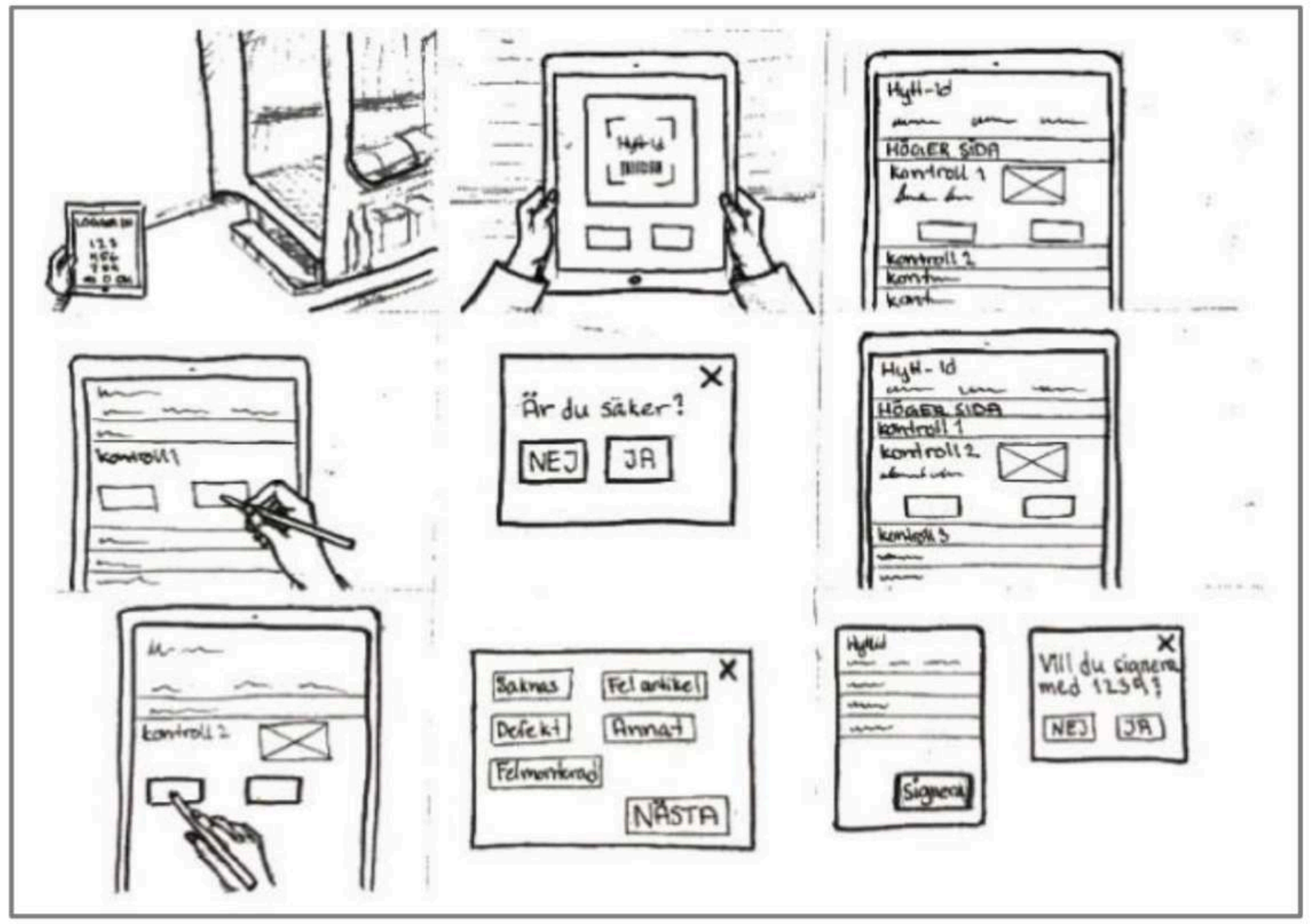


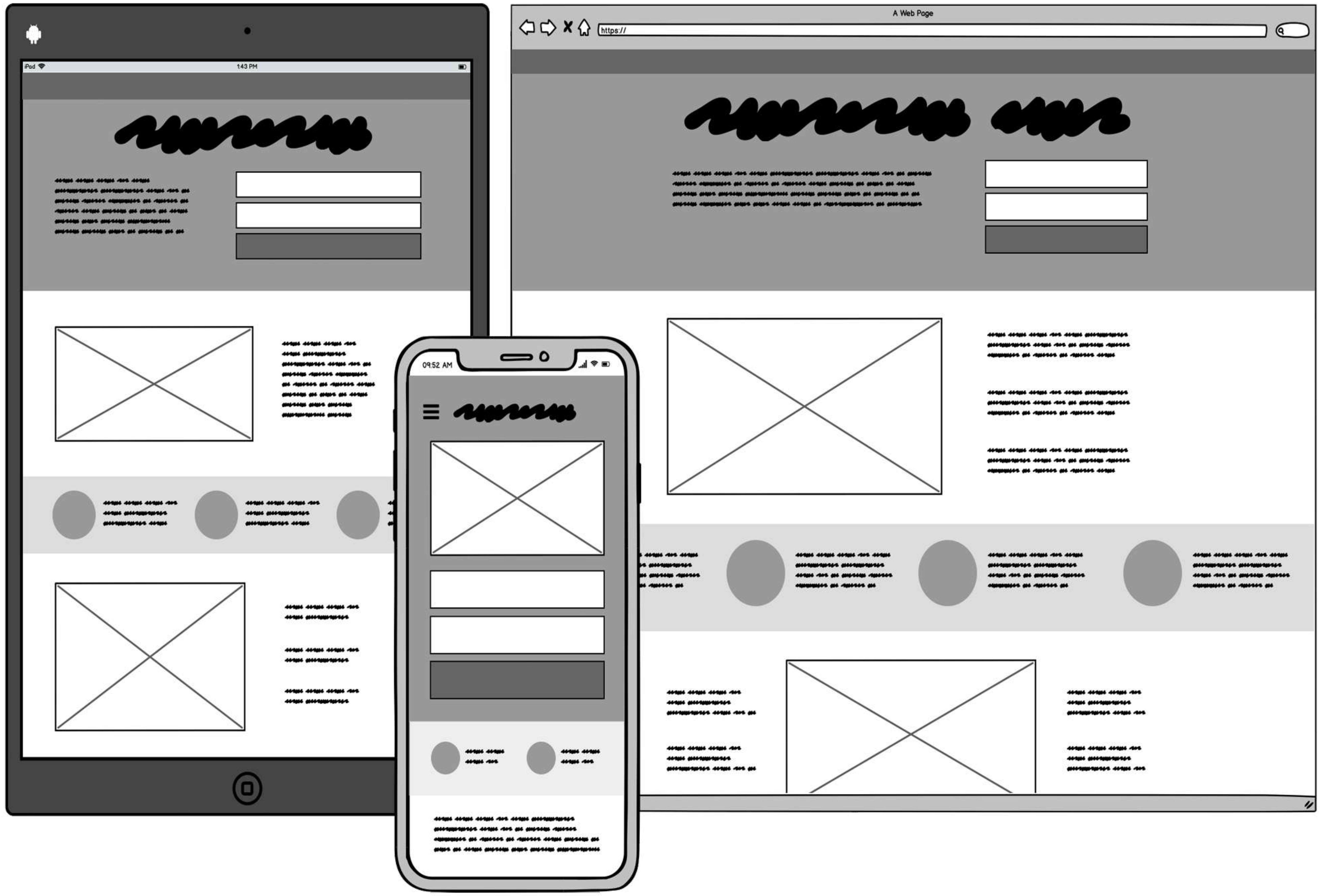
Figure 4. Storyboard that depicts the process of inspecting a cabin with the help of a digital touchpad

Wizard of oz prototyping

A technique that simulates the behavior of a complex system or feature using a human proxy who manually controls the system's responses. It is useful for testing user interactions and system behavior without having to implement the full system.

Wireframe prototyping

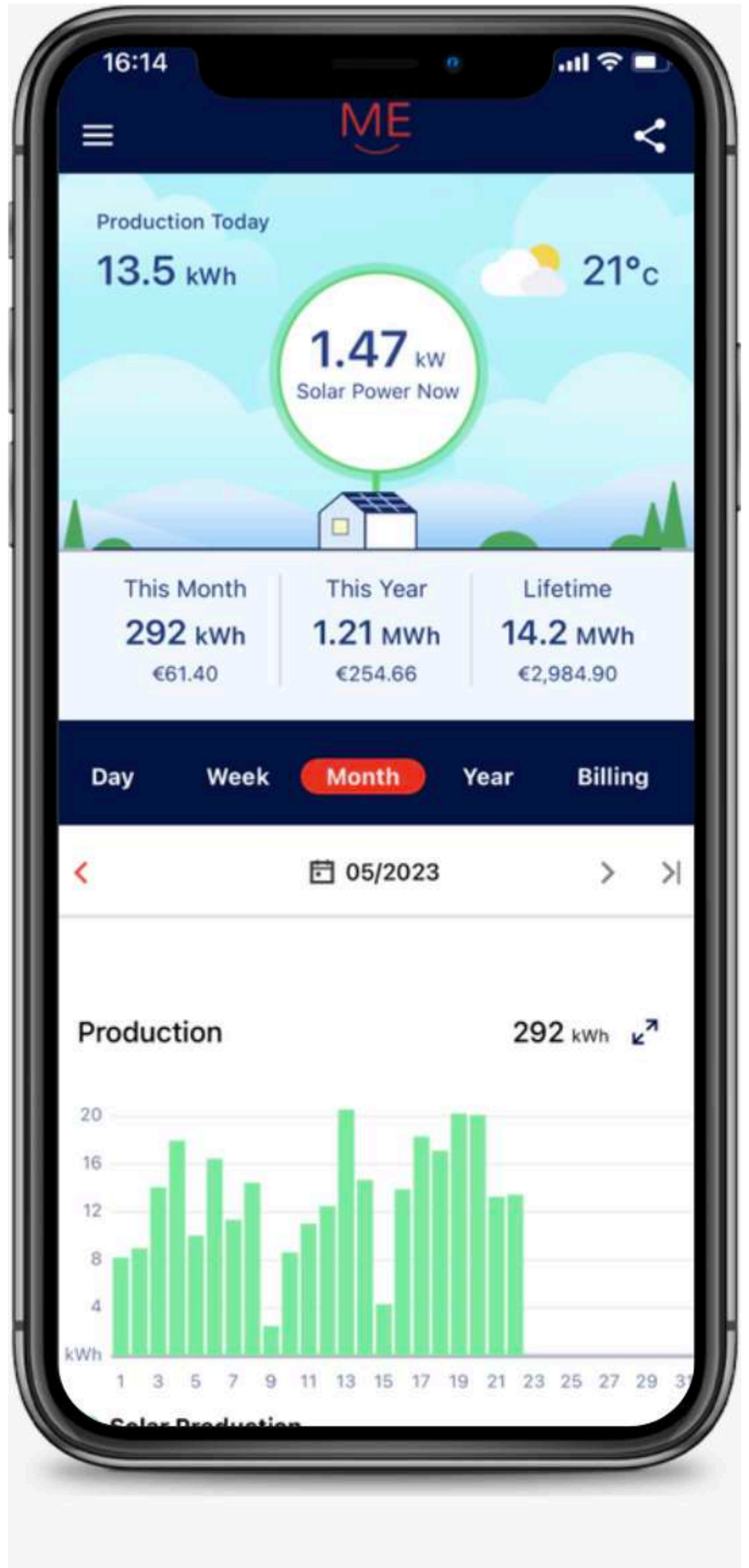
It is a high-level representation of a user interface using simple shapes and lines to outline the layout and placement of the elements. It can be done with pen and paper, presentation software (such as PowerPoint, Canva, Miro) or with specific wireframe software, such as Balsamiq. It is useful to focus on the overall structure and functionality of the interface without getting distracted by details such as fonts, colors and images.



Mockup/Digital Prototyping

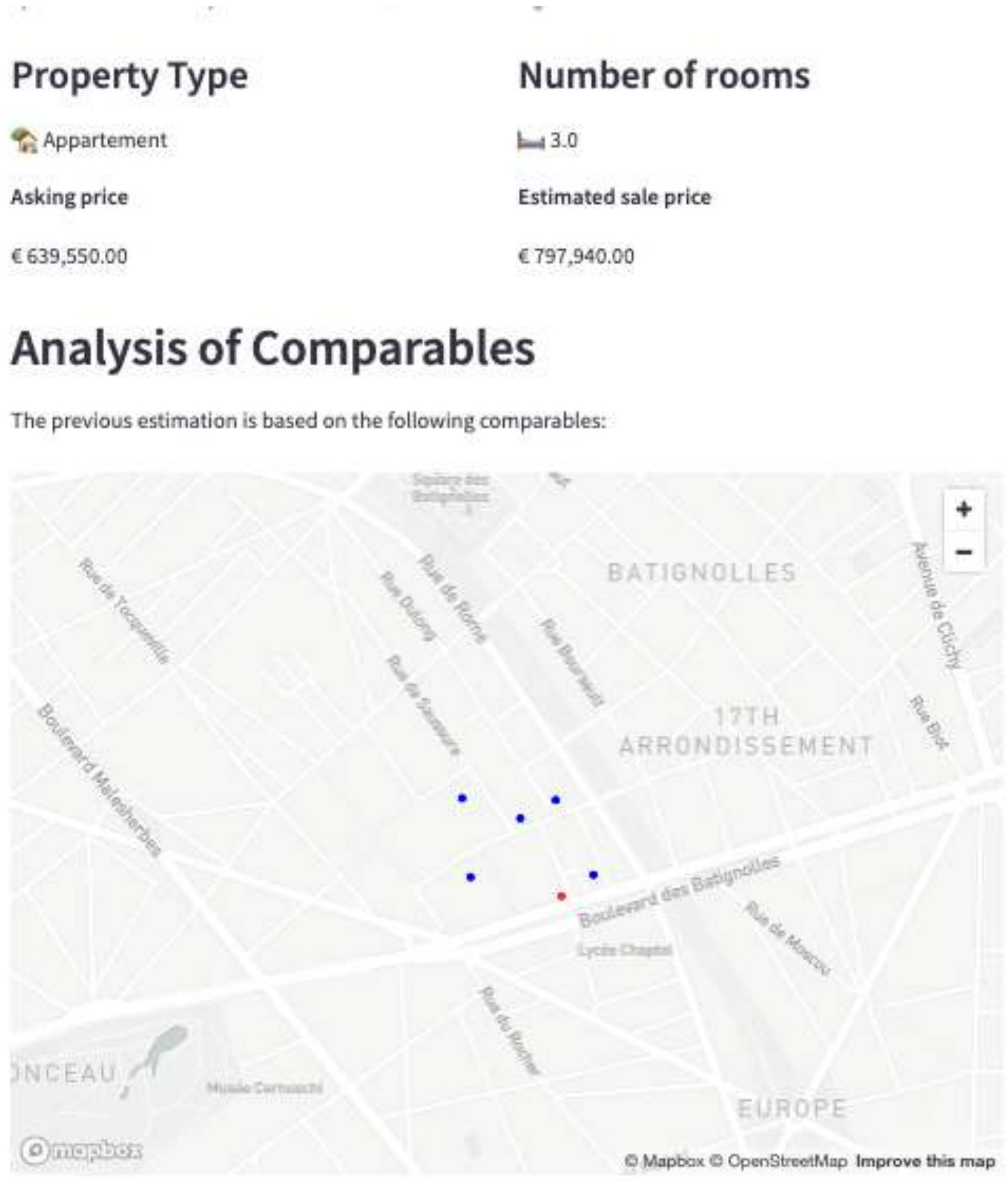
It is a more detailed representation of a user interface than a wireframe, with realistic colors, fonts and icons, and possibly with some interactivity, but still not real software. It is useful when the look and feel is important.

Professional software for digital prototyping includes Principle, Sketch or Marvel.



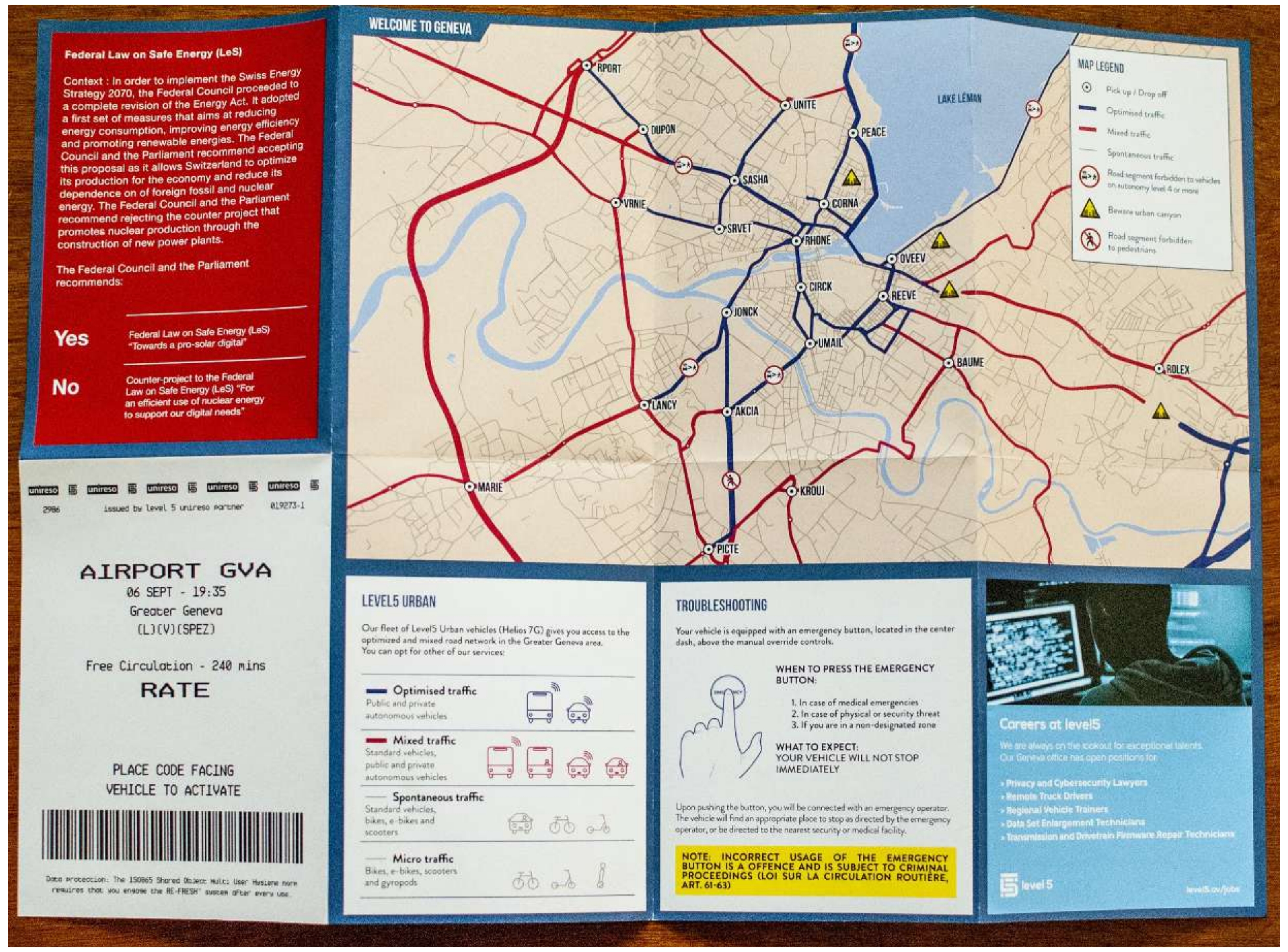
Software prototyping

It is a real piece of software that is more basic than the final product or that implements only a subset of the features. It is useful at later iterations, when it comes to making decisions about the implementation, or when it is important to get the functionality tested.



Design Fiction

Design fiction is the practice of creating tangible and evocative prototypes from possible near futures, to help discover and represent the consequences of technology.



https://nearfuturelaboratory.com/blog/2024/11/geneva-map-for-an-av-future/

3

How does data & AI change all this?

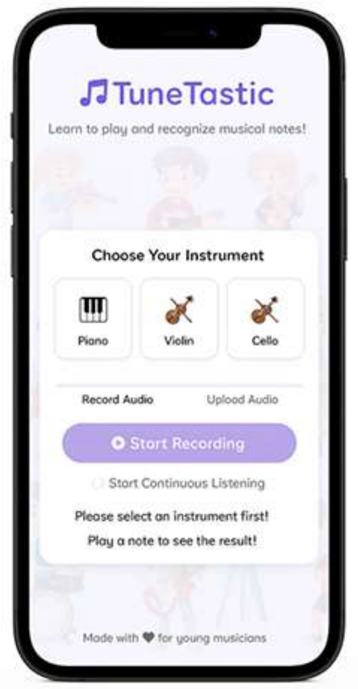
AI is a prototyping tool by itself…

https://dobetter.esade.edu/en/AI-digital-product-prototyping

Prototypes made with ChatGPT, Lovable, Teachable Machine

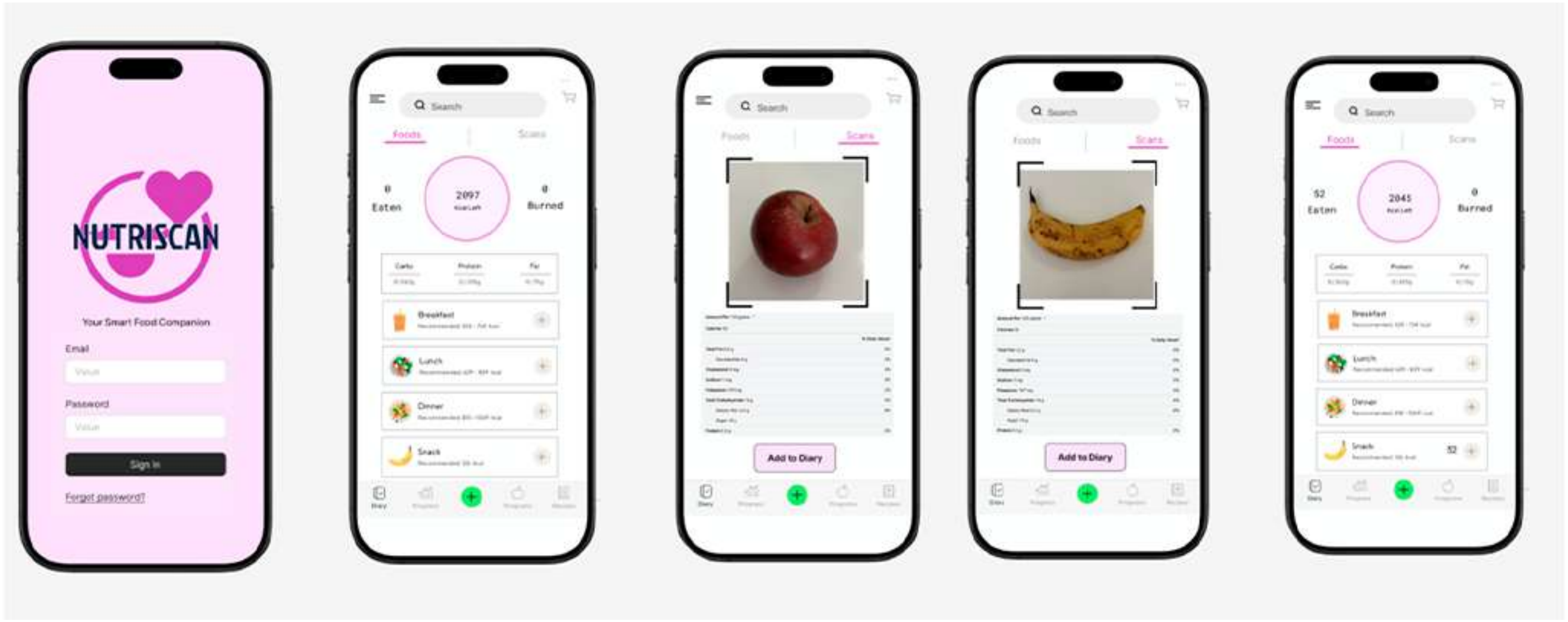
Storyboard

Sketch



Mockup

Wizard of Oz



Design fiction

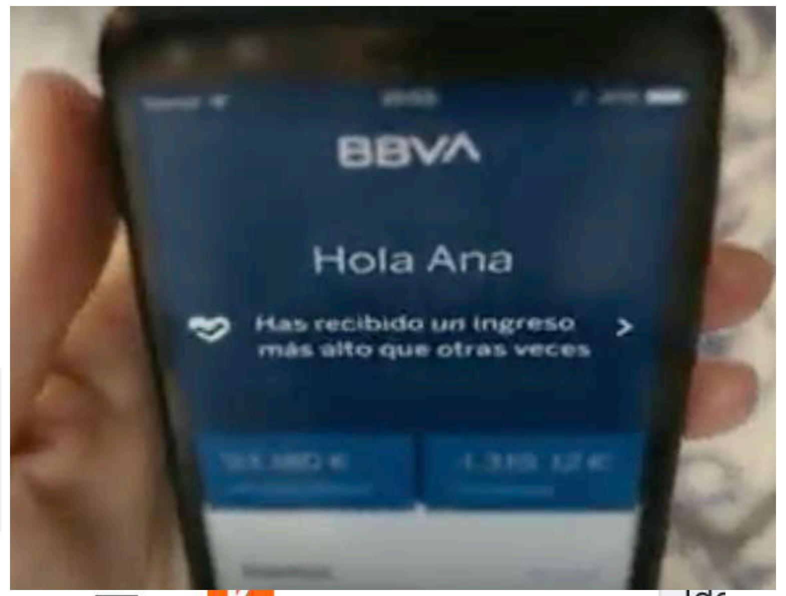
Data and AI are “tactile”

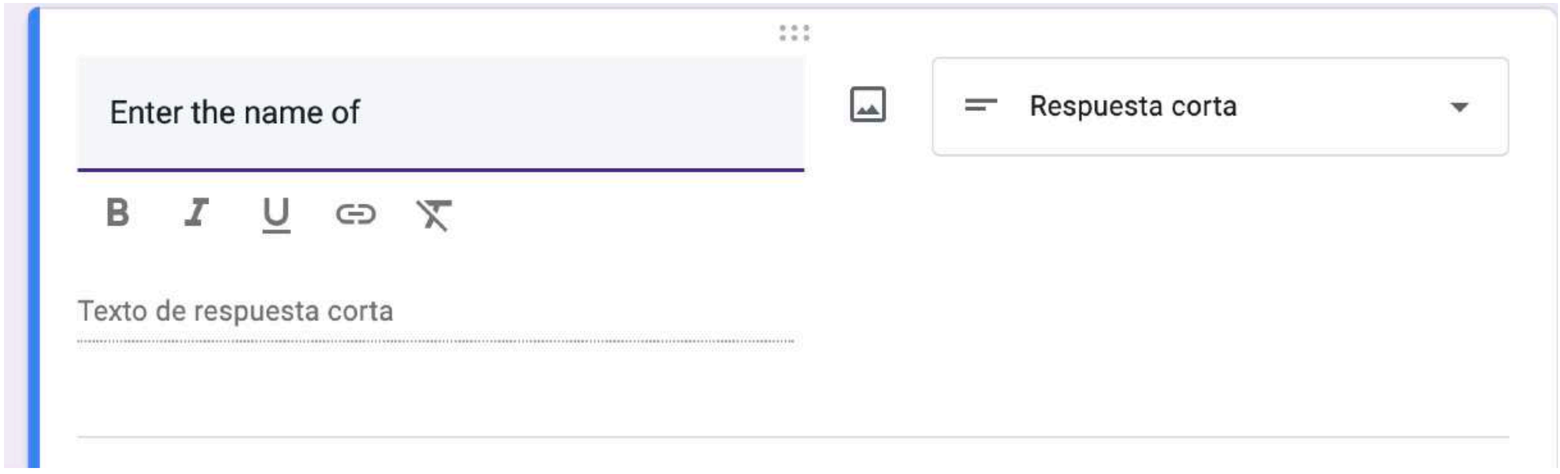
Data and AI change the experience…

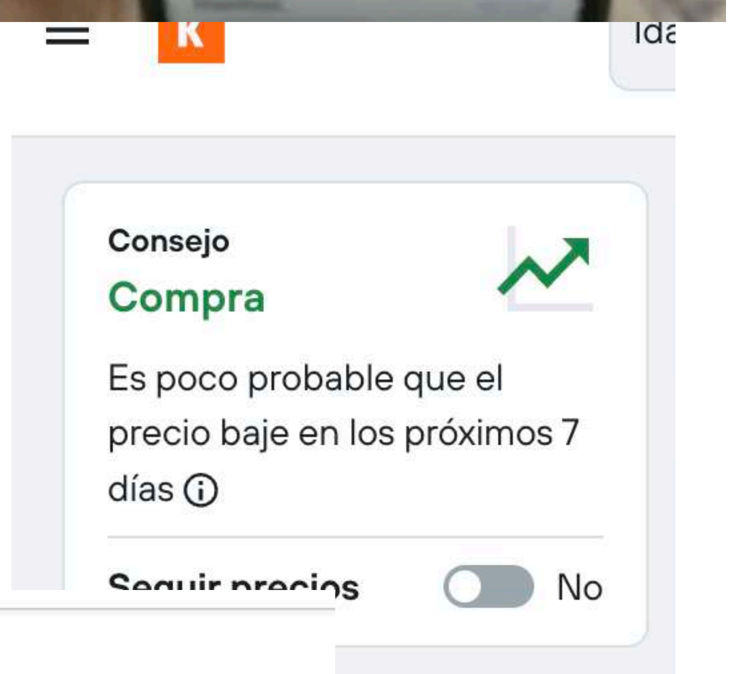
- From “You search for content” to “Content finds you"
- From “shop then ship” to “ship then shop”
- From “Enter information and upload image” to “upload image and autocomplete info”.

… but there’s no way to know without testing.

Plus, you can have a perfect model and get a completely wrong experience.





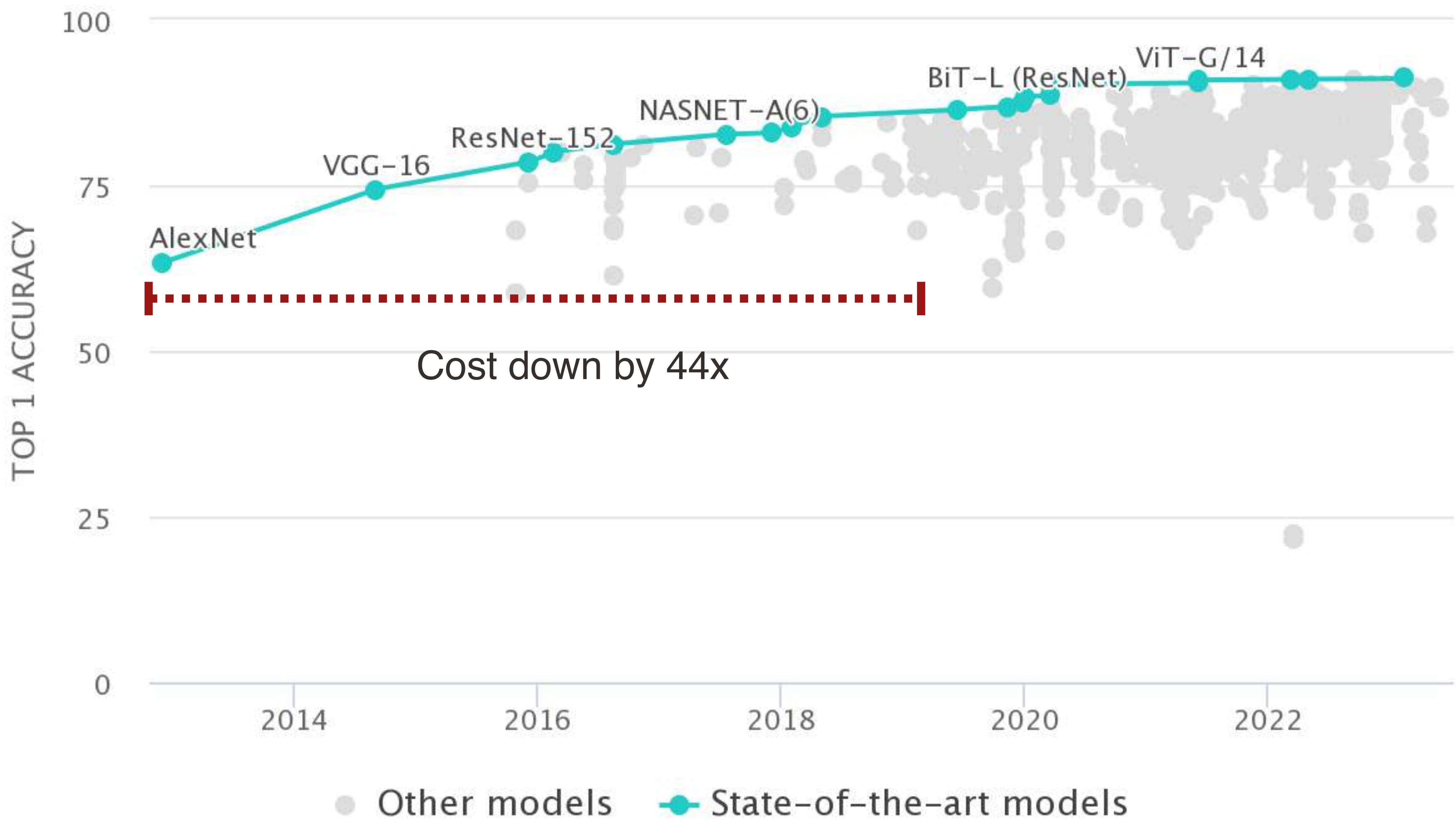


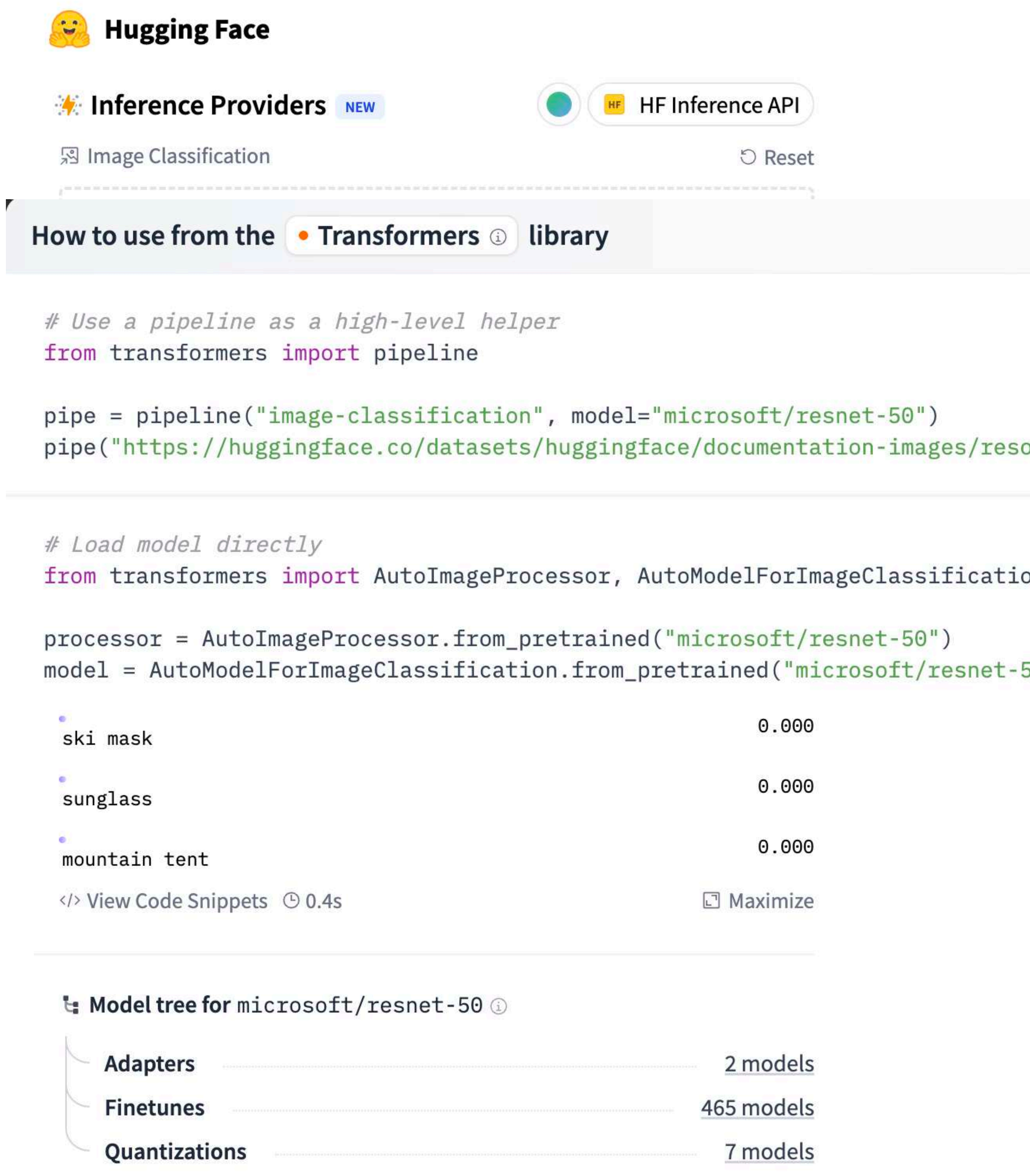
Data / AI Components are more commoditized

Data / AI components are more "commoditised"

- Algorithms to solve AI tasks have become better and cheaper

- Plethora of algorithms and datasets are **readily available** through open-source libraries, cloud services, APIs or model hubs. <u>Incorporating those in prototypes is cheap!</u>





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In 2025, building a real software prototype that uses real data and makes calls to real AI services is easier and faster than ever.

As the cost of AI components goes down, the skill of prototyping with AI will become more valuable.

Do Better.
by esade

The increasingly valuable skill of prototyping with AI

Innovation & technology

Types of AI Products

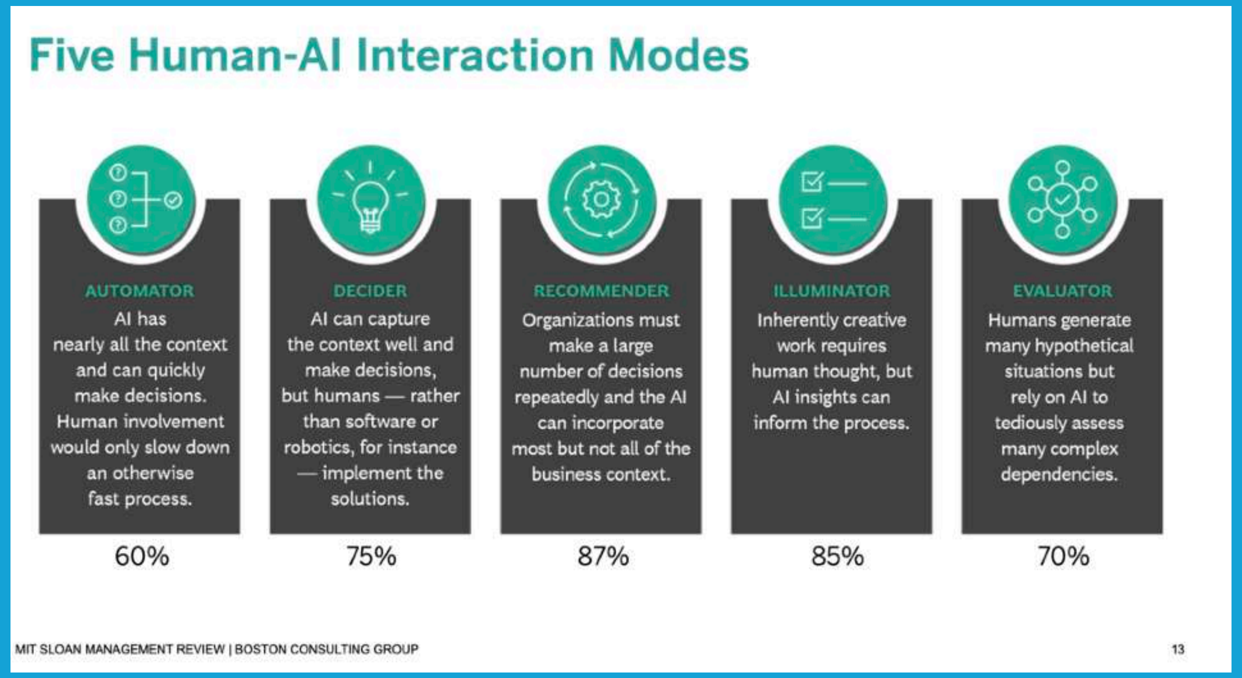
> **Automator**: Fully automate an action more efficiently than humans (e.g. credit card fraud detection, spam detection, security brake).

> **Decider**: Provides a decision but does not have the context or and needs human to take action (e.g. credit concession, medical diagnosis).

> **Recommender**: Provides a recommendation under high uncertainty, but is cheap and valuable for some (e.g. shopping recommendations, job recommendations).

> **Illuminator**: Provides insights that help complementing human understanding (e.g. Fitbit dashboard, customer segmentation).

> **Evaluator**: Provides a complex computation but needs the human to make sense of it (e.g. real estate prediction, anomaly detection).



Learn to Make the Most of Your Relationship with AI, MIT Sloan and BCG

Recap

Data / AI product require a different thinking

- Data collection / quality determines the experience (even if product is great in terms of look and feel and is technically feasible).
- Inherent uncertainty due to non-perfect accuracy.

Data / AI components are more "commoditised"

- Many ready-to-use services available that can be integrated with other pieces of software
- API Revolution

AI is a new tool for prototyping by itself

- Relatively easy to generate ideas* and visual materials with Gen AI
- Relatively easy to generate websites, python code scripts
- New "no code" is coding with GenAI. Assistants like Lovable, Bolt on focus.

You need to think in AI terms

You can include real data / AI in your prototypes (both software and non-software ones)

You can use GenAI to generate materials, sketches, concepts, even the code of a website or a program.

5

Hands-on activity

Activity

You are a data scientist at LinkedIn. The company is considering to add a "Books you may like" feature. You are asked to explore how to add book recommendations for a user given their Activity (timeline).

> Make groups of 2-3 people with your neighbours.

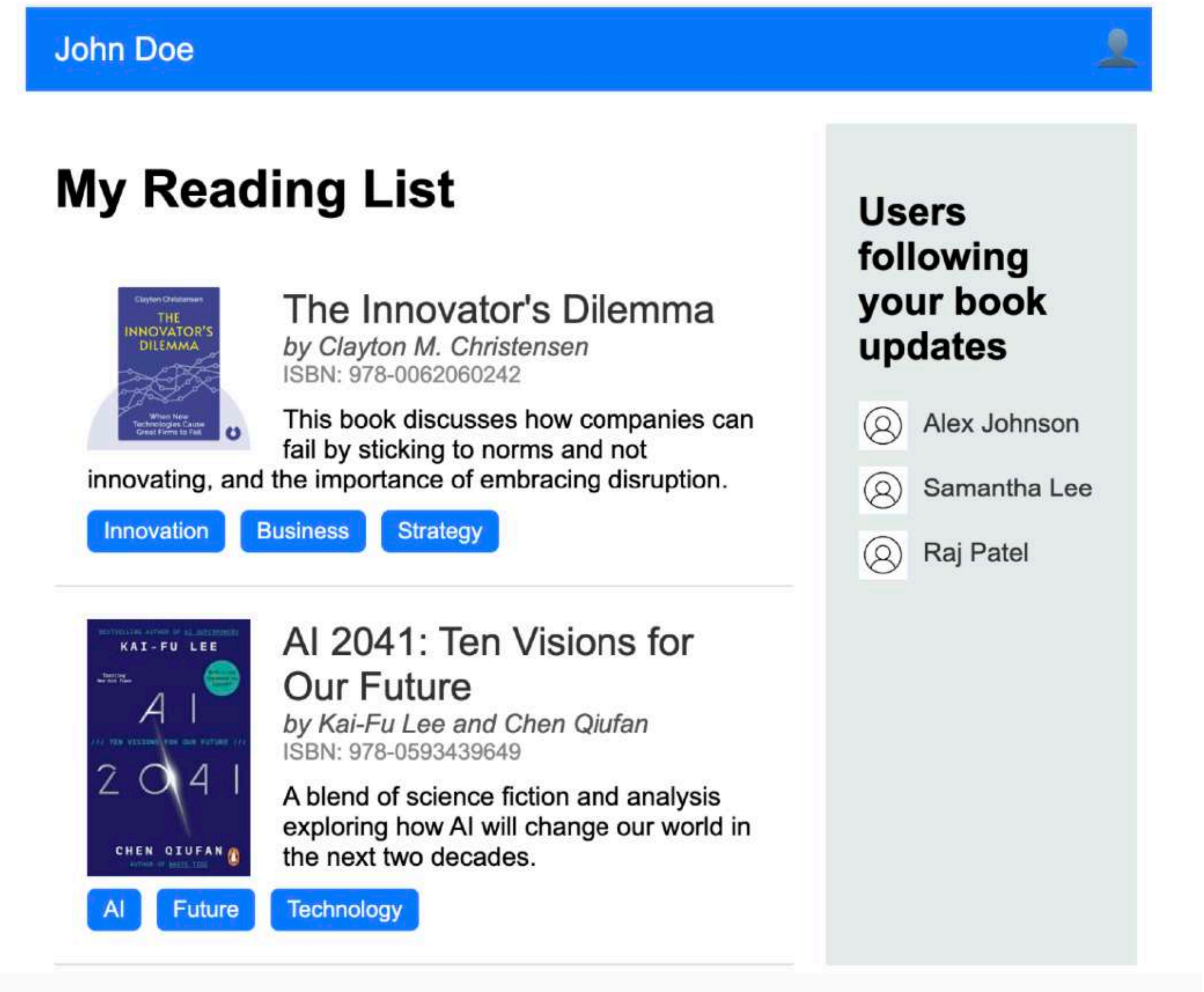
> Brainstorm ideas on the technical approach you would follow.

> Brainstorm ideas on how you can develop a quick "demonstrator" of your chosen approach.

> Try to implement the demonstrator with any means and knowledge you have at your disposal (any prototyping methods discussed, generative AI, simulated data, real data, python…).

> You can focus on functionality, on appearance, or both.

> Try to follow the general philosophy discussed in this class (low fidelity, features of prototypes…).



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